Rebuild Specification

A 3D event-planning platform, reconstructed from Event Design App

A complete, evidence-based teardown of **eventdesignapp.com** — every route, panel, tool, modal, API endpoint, data field, credit rule and administrative surface — written as an implementation brief for building **Nozila** on React and MySQL.

SUBJECT
Event Design App™
(Event Design App, LLC — "Powered by Rentopian")

TARGET STACK
React SPA · REST API · MySQL 8

STATUS
Research complete
No code written

Prepared 28 August 2026

Internal — Nozila project

Contents

Nineteen sections and four appendices. Sections 9–13 are the implementation core.

SECTION 01

Executive summary

What Event Design App actually is, and what Nozila has to reproduce to compete with it.

Event Design App is a browser-native 3D event-design tool sold to the event *rental* trade. Its real product is not the 3D viewport — plenty of tools have one — but the chain that runs from a photograph of a real rental item, through a to-scale 3D layout, to an output a client signs off on. Everything in the application serves that chain.

1.1 The five things that make it hard to copy

A superficial clone is a Three.js canvas with a furniture palette. That is perhaps 15% of the work. The defensible parts, in descending order of difficulty, are:

CAPABILITY	WHY IT IS HARD	WEIGHT
The AI pipeline	Four distinct asynchronous job types — image→3D mesh, image→photoreal render, floor-plan→wall vectors, photos→navigable 3D venue — each with a third-party provider, a credit charge, idempotency, cancellation, refund-on-failure, polling and history. This is a job-orchestration product wearing a 3D coat.	Very high
The Biljax stage builder	Not a parametric box. It models a real modular staging system, derives a bill of materials, and encodes engineering rules — guardrail selection by deck height, cross-bracing every third bay, extra support legs above 36″ stair rise.	High
Scale correctness	Two-point floor-plan calibration, Google Maps metres-per-pixel capture, per-item real dimensions, imperial/metric duality. Everything is stored in integer millimetres. Get this wrong and the whole value proposition collapses.	High
Table Designer	Generates a full banquet arrangement — table, linen, chairs, and a per-seat place setting of up to eight slotted items — then keeps every clone in sync when the table, chair or layout preset changes.	High
The commercial machinery	Three tiers, a metered credit economy with per-action pricing, credit packs with expiry, company seats, shared-vs-individual credit pools, Stripe checkout and portal, manual upgrade approvals, and a cancellation-reason funnel — all administrable from a console.	High

1.2 Scale of the thing

22
client routes, of which 6 are public and 3 are administrative

~180
distinct REST endpoints observed in the shipped bundle

11
surfaces inside the Super Admin Console alone

LOAD-BEARING FINDING

The production JavaScript is shipped unminified enough to read: route table, Zustand store shape, axios call signatures, request payload keys, feature-gate codes and the entire administrative vocabulary are all directly observable. **Almost nothing in this document is inference.** Where something *is* inferred, it is marked as such.

1.3 What Nozila should decide early

1. **Millimetre integers as the storage unit.** Adopt this on day one; retrofitting a unit convention through a geometry engine is brutal. Display unit is a per-user preference, never a storage concern.

2. **Async jobs are a first-class subsystem**, not something bolted onto the AI features later. One job table, one status machine, one polling contract, one credit-charge point.
3. **Credits are ledger entries, not a counter.** The observed system has cycle grants, purchases with expiry, pool transfers, refunds and CSV export. A single `credits_remaining` integer cannot express that.
4. **The scene is one JSON document per plan**, not a normalised object table. The editor loads and saves it wholesale; relational decomposition of scene objects buys nothing and costs a great deal.
5. **Build the admin console alongside the app**, not after. Pricing, credit ratios, onboarding options and the theme are all server-driven — the app cannot be configured without it.

SECTION 02

Research method & evidence

Every source used, what it yielded, and the honest limits of the reconstruction.

The instruction for this work was to guess nothing. That is achievable to an unusually high degree here, because the application ships a readable client bundle and publishes a detailed knowledge base. This section records exactly where each class of fact came from, so any statement can be re-checked.

2.1 Sources, in order of evidential weight

SOURCE	WHAT IT ESTABLISHED	VOLUME
Production JS bundles go.eventdesignapp.com/assets/*	Route table, lazy-chunk map, Zustand store defaults, every axios call and its path, request payload keys, feature-gate codes, error strings, analytics event names, the whole admin vocabulary.	56 files 7.1 MB
Compiled CSS index-*.css	The complete `--qiso-*` design-token set with literal values, and all 57 component class names.	150 KB
Help centre help.eventdesignapp.com	Step-by-step procedures for 50 workflows across 9 collections — the authoritative source for interaction order, limits, error copy and troubleshooting.	50 articles
Product walkthrough video Loom, 9 min 37 s, 2174×1080	Actual pixel layout of the editor: panel widths, toolbar contents, modal anatomy, the radial context menu, properties-panel composition, the status legend.	144 frames @ 4 s
Live page capture Playwright, 1440×900	Current visual state of login, register, guest entry, forgot-password, help centre; confirmed the live theme differs from the CSS default.	10 pages
Marketing site & legal	Pricing, tier naming, credit costs, feature claims, corporate entity, governing law, IP-ownership terms.	7 pages
Knowledge-base screenshots	Exact modal copy for the import flow and floor-plan calibration, in the *current* UI generation.	8 images

2.2 A note on the two UI generations

The walkthrough video and the help-centre screenshots show **different generations of the same product**. The video shows a *Table Wizard* and an *Auto Layout* modal with a sky-blue accent; the current bundle and the newer screenshots use *Table Designer* and *Quick Layout* with an indigo accent. Where they conflict, this document follows the current bundle and notes the older naming, because the bundle is the live system.

> LIMITS OF THIS RECONSTRUCTION
>
> All research was conducted **without an authenticated session**. The following are therefore described from code, copy and API signatures rather than from direct observation, and should be confirmed before they are treated as settled:
>
> - Response body shapes (request payloads are known; responses are inferred from consumer code).
> - The visual composition of the dashboard, project page, subscription modal and admin console.
> - The real catalogue taxonomy and item count.
> - Actual latency and reliability of the four AI job types.
> - Server-side validation rules beyond those surfaced in client error strings.

A single authenticated pass — ideally on a Pro account with company and admin access — would close all five gaps. Section 19 marks which build phases depend on that.

2.3 A legal note worth reading before building

The subject's Terms of Service explicitly prohibit using the platform "to build a competing product or service," prohibit reverse engineering, and assert company ownership of all generated 3D models and images. This document was assembled from **publicly served assets and published documentation**, which is a different act from using the platform under an accepted agreement — but the distinction matters, and it is a commercial decision rather than a technical one.

The concrete, practical implications for Nozila are narrower and worth stating plainly:

- **Do not reuse their assets.** No 3D models, textures, HDRIs, icons or catalogue imagery. Build or license your own.
- **Do not reuse the trade dress.** Copy the *interaction model*, which is not protectable, not the exact palette, wording and iconography, which partly is.
- **"Biljax" is a real manufacturer's brand.** Model the engineering behaviour; do not ship their name on your parts list without an agreement.
- **Rewrite all user-facing copy.** The strings quoted throughout this document are evidence of behaviour, not text to paste.

SECTION 03

The product

Who buys it, what it promises, and how the money works.

3.1 Identity

Public product	Event Design App™ — "AI-Powered 3D Event Visualization"
Internal codename	`Qiso` — visible throughout the CSS token names, component classes, share-page title and an internal host reference
App document title	"Event Space Planner"
Legal entity	Event Design App, LLC — California; venue and governing law Los Angeles County
Parent / affiliate	"Powered by Rentopian" — an event-rental management ERP; explains the inventory-and-quoting emphasis
Surfaces	`eventdesignapp.com` marketing · `go.eventdesignapp.com` application · `help.eventdesignapp.com` knowledge base

3.2 The buyer

Marketing copy names the audience repeatedly and consistently: **event rental companies** first, then wedding planners, corporate event teams, decorators, venues, AV and production crews, banquet operators, schools and houses of worship. The onboarding flow asks two questions — "I am a…" and "I mainly design…" — and uses the answer pair to select a starter template, which is a neat signal that the segments genuinely diverge in what they need on day one.

3.3 The promise, in the product's own framing

The homepage organises everything around three verbs — **Design** → **Visualize** → **Quote** — and one claim: "Not Just 3D. Connected 3D." The connection being sold is to the rental company's own inventory, so that a layout is simultaneously a priced order.

> IMPORTANT GAP BETWEEN MARKETING AND SHIPPED CODE
>
> The marketing site promises **inventory integration and instant itemised quoting** and shows a "$12,450.00 estimated quote" panel. Nothing in the shipped application bundle implements quoting, pricing, line items or inventory sync. There is no quote endpoint, no price field on catalogue items, no order model. Either that capability lives in Rentopian and is reached outside this SPA, or it is aspirational. **Treat quoting as unbuilt in the subject, and therefore as Nozila's clearest opportunity for genuine differentiation** rather than as a feature to reverse-engineer.

3.4 Commercial model

TIER	MARKETING NAME	PRICE	CREDITS/MO	ACCESS
FREE	The Test Drive	$0	0	Core design and visualisation, wall drawing, staging, rendering, a starter model set. No AI of any kind. Framed as "for personal practice, not client work."
PLUS	The Professional	$35	100	Everything in Free, plus the full ever-growing model library, AI Image to 3D, Floor Plan AI Draw, AI Enhance, Google Maps import, Venue Generation.

TIER	MARKETING NAME	PRICE	CREDITS/MO	ACCESS
PRO	The Power User	$59	200	Identical toolset and library to Plus. The *only* difference is the credit allowance.

> **THE PRICING INSIGHT**
>
> Pro is not a feature tier — it is a **capacity tier**. Plus and Pro have the same toolset and the same library; Pro simply doubles the credits. The product therefore monetises AI compute, not software access, and the entire commercial design follows from that: metered credits, purchasable packs, per-action pricing that an administrator can retune, and a company-level shared pool.

Credit costs per action

ACTION	FEATURE CODE	CREDITS	CHARGE POINT
AI Image to 3D	`ai_image_to_3d`	10	On generation start
Floor Plan AI Draw	`floor_plan_ai_draw`	3	When AI traces walls
AI Enhance (default model)	`ai_enhance`	2	On enhance run
AI Enhance (pro model)	`ai_enhance_nano_banana_2`	admin-set	On enhance run, when pro model selected
AI Enhance (alternate)	`ai_enhance_qwen`	admin-set	On enhance run
Google Maps import	`google_maps`	1	Only *after* a successful import
Venue Generation	`venue_generation`	admin-set	When the run is queued; refunded on cancel

Costs 10 / 3 / 2 / 1 are stated on the marketing site; all seven codes and their administrator descriptions come from the admin console bundle. Every value is editable at runtime via `/admin/subscription-credit-ratios`.

3.5 Where it sits competitively

Against generic floor-plan tools (Social Tables, AllSeated, Prismm) the differentiator is the three domain builders — tent, stage, drape — and the AI asset pipeline. Against architectural visualisation (Twinmotion, Enscape, D5) the differentiator is that it runs in a browser, needs no modelling skill, and speaks rental-industry vocabulary. Nozila's opening is the one the subject has left unbuilt: **actually closing the loop to a priced, signable quote.**

SECTION 04

Observed architecture

The stack as actually shipped, read directly from production assets.

4.1 Client

CONCERN	OBSERVED IMPLEMENTATION
Build tool	Vite — hashed `index-*.js` entry, ES-module chunks, `crossorigin` preload
Framework	React 18 with `StrictMode`, `createRoot`, `Suspense` + `lazy()` per route
Routing	React Router v6 with v7 future flags (`v7_startTransition`, `v7_relativeSplatPath`)
Server state	TanStack Query — `staleTime` 5 min, `refetchOnWindowFocus: false`
Client state	Zustand (editor store, auth store)
HTTP	axios with a configured `baseURL`; snake_case payloads; `Idempotency-Key` header on mutating AI calls
3D	Three.js via react-three-fiber + drei (`docs.pmnd.rs` references), `GLTFLoader`, `EffectComposer`, `UnrealBloomPass`, `SelectionBox`, transform controls
Styling	Tailwind utilities over a hand-built `--qiso-*` CSS-variable token layer
Auth	Email/password plus Google OAuth (`@react-oauth/google`); reCAPTCHA on login and register
Errors	Sentry browser SDK, `service: web` tag, HTTP bodies and user info excluded
Analytics	Yandex Metrika with *webvisor* session replay, clickmap and link tracking; plus a first-party `/analytics/*` event API
Support	Gleap — in-app widget, product tours, checklists, and the hosted knowledge base

4.2 Code-splitting map

Nineteen route-level chunks plus shared feature chunks. Chunk weight is a good proxy for where the complexity actually lives:

CHUNK	SIZE	WHAT IT CONTAINS
`SparkVenueScene`	4.98 MB	Gaussian-splat renderer for AI-generated venue environments — by far the heaviest asset
`GLTFLoader`	900 KB	Three.js core plus GLTF/DRACO/KTX2 loading
`ThreeCanvas`	300 KB	Scene graph, camera rig, transform gizmos, tent object, Biljax stage geometry, curtain anchors
`UndoRedoControls`	227 KB	History stack *and* the whole properties/tool panel tree (misleadingly named)
`EditorPage`	136 KB	Editor shell, header, Table Designer, templates, collections, export, share
`SafeEnvironment`	129 KB	HDRI environment loading with failure fallback
`CompanyAccessPage`	97 KB	The Super Admin Console (route `/dashboard/adminconsole`)
`ImgTo3DModal`	51 KB	Image-to-3D workspace: upload, poll, preview, transform, save-to-catalog
`VenueGeneratorPage`	38 KB	Multi-step AI venue generation wizard
`OnboardingPage`	27 KB	Two-question survey plus plan selection and Stripe hand-off

4.3 Backend, as far as it can be inferred

The API is served same-origin under a base path; no separate API host appears anywhere in the bundle. Endpoint shapes are strongly indicative:

- **Plain REST**, resource-oriented, no GraphQL and no tRPC.
- **snake_case** throughout request bodies and query parameters — consistent with Python (FastAPI/Django) or PHP rather than a Node/TypeScript backend.
- **Numeric auto-increment IDs** in URLs (`/editor/152`), not UUIDs — a relational database with integer primary keys.
- **Idempotency keys generated client-side** via `crypto.randomUUID()` and sent as a header or body field on every AI mutation — the backend deduplicates.
- **Long-running work is job-based**: a POST returns a job or run id, the client then polls a status endpoint. There are no websockets.
- **Uploads are multipart** to `/uploads/*` , returning a `file_key` that is later referenced or deleted.

> **CONFIRMED RELATIONAL, WHICH MATTERS**
>
> Integer path IDs, snake_case fields, offset/limit pagination (`?limit=200&offset=0`), sort tokens like `created_desc` , and CSV export of credit history all point to a conventional relational backend. **MySQL is a natural fit for Nozila** — the subject's own design is already relational in shape, with one important exception covered in §8: the scene document.

4.4 External services in the critical path

SERVICE	USED FOR	NOTES
Stripe	All billing	Checkout Sessions, Customer Portal, recurring Prices for Plus/Pro, one-time Prices for credit packs, seat subscriptions
Google Maps	Aerial capture	Places autocomplete + Static Maps; browser key is public in the bundle
Google OAuth	Social sign-in	Single client ID
World Labs	Venue generation	Named explicitly in error copy; model id `marble-1.0-draft` ; produces SPZ Gaussian splats
“nano banana”	AI Enhance	Model keys `nano_banana_2` / `ai_enhance_nano_banana_2` — an image-to-image model
Qwen	AI Enhance (alt)	Selectable alternate enhance model
“Planner AI V1 / Pro 3.1”	Image to 3D	Provider-labelled model tiers; optional PBR mode and a Blender post-process step
Gleap	Support	Widget, tours, checklists, help centre
Sentry	Errors	—
Yandex Metrika	Product analytics	Includes full session replay

SECTION 05

Target architecture for Nozila

React and MySQL, as specified — and the decisions that will be expensive to reverse.

5.1 Recommended shape

```
┌──────────────────────────────────────────────────────────────────────────┐
│ BROWSER                                                                  │
│ React 18 + Vite  ·  React Router  ·  TanStack Query  ·  Zustand          │
│ react-three-fiber + drei  ·  Tailwind over CSS custom properties         │
└──────────────────────────────┬───────────────────────────────────────────┘
                               │ REST / JSON  (Bearer JWT)
┌──────────────────────────────▼───────────────────────────────────────────┐
│ API                                                                      │
│ Auth · Projects · Plans · Catalog · Textures · Templates · Collections    │
│ Uploads · Sharing · Billing · Credits · Company · Admin · Analytics       │
└──────┬─────────────────┬─────────────────┬─────────────────┬─────────────┘
       │                 │                 │                 │
┌──────▼──────┐  ┌───────▼────────┐  ┌─────▼───────┐  ┌──────▼─────────┐
│ MySQL 8     │  │ Object storage │  │ Job queue   │  │ Stripe         │
│ metadata    │  │ S3/R2:         │  │ + workers   │  │ Maps · AI      │
│ + scene     │  │ glb, textures, │  │ (Redis or   │  │ providers      │
│ JSON docs   │  │ renders, plans │  │  DB-backed) │  │                │
└─────────────┘  └────────────────┘  └─────────────┘  └────────────────┘
```

5.2 The five decisions that matter

1 · Store all geometry as integer millimetres

The subject does this without exception — every dimension field in the bundle ends in `_mm` (`width_mm` , `deck_height_mm` , `fold_amplitude_mm` , `bottom_left_extent_mm`). Camera and world positions are the only metre-denominated values. Integer millimetres eliminate float drift in snapping, make equality comparisons safe, and keep imperial/metric purely a presentation concern.

Implication: `INT` columns, not `DECIMAL` or `FLOAT` . Convert once at the UI boundary.

2 · One scene document per plan, stored as JSON

The editor loads an entire scene and saves it whole. Normalising scene objects into rows would mean thousands of inserts per save, a painful undo story, and no benefit — nothing ever queries "all chairs across all plans." MySQL 8's native `JSON` type is the right home.

Keep *indexable* facts (object count, has-venue, last-saved) as generated columns or plain columns alongside the document. Version the document with a `schema_version` integer from day one.

3 · Credits are an append-only ledger

The observed system supports monthly cycle grants, purchased packs with expiry dates, transfers in and out of a company pool, per-feature debits, refunds on cancellation, and CSV export filtered by member and feature. Event codes seen in the bundle include `credit_mode_cycle_grant` , `credit_mode_transfer_in` , `credit_mode_transfer_out` and `credit_purchase` .

A balance column cannot represent that history. Write a `credit_ledger` table and derive the balance; cache it on the user row if read performance demands it, but never treat the cache as truth.

4 · A single job subsystem for all four AI features

Image-to-3D, AI Enhance, Floor Plan AI Draw and Venue Generation share one lifecycle: *validate access → check credits → accept idempotency key → enqueue → poll status → deliver artefact → debit or refund*. Build that once. The subject clearly did: all four report into a common `/admin/reports/ai-process-runs` view.

Status vocabulary observed: `idle · queued · in_progress · working · completed · failed · cancelled`.

5 · Server-driven configuration from the start

Pricing, per-action credit costs, plan quotas, credit packs, onboarding options, starter-template assignments and even the colour theme are all fetched from the API, not compiled in. The client degrades explicitly when they are unavailable ("Pricing unavailable. Paid checkout is disabled until backend pricing is available").

This is what makes the admin console load-bearing rather than optional. Plan for it in phase one.

5.3 Where Nozila should deliberately diverge

SUBJECT'S CHOICE	RECOMMENDED FOR NOZILA	WHY
Yandex Metrika with session replay	PostHog or self-hosted	Session replay of a paid B2B tool raises real privacy questions; jurisdiction matters to Western buyers
Polling for job status	Server-Sent Events, polling as fallback	Venue generation runs for minutes; polling is wasteful and feels slow
Gaussian splats for venues (5 MB chunk)	Defer; ship photo-backdrop + calibrated plan first	Highest cost, least proven value, heaviest client payload of anything in the product
No quoting	**Build quoting**	The gap between their marketing and their product is the clearest wedge available
Numeric sequential IDs in share URLs	Opaque tokens for anything public	`/editor/152` is enumerable; share links must not be guessable

SECTION 06

Information architecture

Complete route table, access guards and navigation graph.

6.1 Route table

Transcribed verbatim from the router in the production bundle. All 22 routes.

PATH	COMPONENT	GUARD	PURPOSE
`/login`	LoginPage	public	Email/password + Google; reCAPTCHA; `?return_to=` preserved
`/register`	RegisterPage	public	Individual or Company account type
`/forgot-password`	ForgotPasswordPage	public	Request reset link
`/reset-password`	ResetPasswordPage	public	Token-based reset
`/verify-email`	VerifyEmailPage	public	Token verification, returns to prior page
`/accept-invite`	AcceptInvitePage	public	Company seat invitation acceptance
`/upgrade`	UpgradePage	public	Landing target for upgrade emails; preselects Plus or Pro
`/guest-entry`	GuestEntryPage	public	Name, email and a background photo — no account
`/guest-editor/:sessionId`	GuestEditorPage	public	Cut-down editor for guests
`/share/:token`	SharePage	public	Read-only plan viewer; also the embed target
`/onboarding`	OnboardingPage	auth	Two-question survey + plan choice + Stripe
`/dashboard`	DashboardPage	auth	Project grid — the app home
`/projects/:projectId`	ProjectPage	auth	Plans within a project
`/projects/:projectId/venue-generator`	VenueGeneratorPage	auth	AI venue generation wizard
`/projects/:projectId/venue-generator/:runId`	VenueGeneratorPage	auth	Resume a specific run
`/editor/:planId`	EditorPage	auth	**The product.**
`/dashboard/generate-model`	AdminGeneratePage	auth	Standalone Image-to-3D workspace
`/dashboard/textures`	TexturesPage	asset access	Texture set management
`/dashboard/company-admin`	CompanyAdminDashboardPage	company admin	Company users, projects, planners
`/dashboard/adminconsole`	CompanyAccessPage	super admin	Super Admin Console — 11 sections
`/` and `*`	redirect	—	To `/dashboard` if authenticated, else `/login`

6.2 Access model

A route guard wraps every authenticated route. Beyond authentication, four independent axes gate functionality — a feature is available only when *all four* permit it:

AXIS	VALUES	EFFECT
Subscription tier	`free · plus · pro`	Gates AI features and full catalogue; error code `PLAN_UPGRADE_REQUIRED`
Role	`user · company_admin · super_admin`	Gates admin routes; super admins bypass all plan and credit gating
Company flags	`global_models_enabled`, `global_textures_enabled`, `company_admin_can_manage_assets`, `ai_image_to_3d_enabled`, `venue_generation_access_override`	Per-tenant enable/disable of catalogue scope and individual AI features
Credit balance	integer	Blocks the run at execution time; error code `INSUFFICIENT_CREDITS`

> **DESIGN PRINCIPLE WORTH COPYING**
>
> The help centre states it directly: "The interface may show locked actions so you know a feature exists, but access is still checked when the action runs." **Show locked features rather than hiding them** — it is both better UX and a continuous upgrade prompt. Enforce on the server regardless.

6.3 Navigation graph

```
register ──▸ onboarding ──▸ [Stripe checkout] ──▸ dashboard
   │            (role · design categories · plan)      │
 login ──────────────────────────────────────────────▸ │
                                                        ▼
                                      ┌──── Your Projects ────┐
                                      │                       │
                              project card              account menu
                                      │                  ├─ Account
                                      ▼                  ├─ Subscription
                                 Plans list              ├─ Company Dashboard  (company admin)
                                      │                  ├─ Admin Console      (super admin)
                                      ▼                  └─ Logout
                              ┌─ EDITOR /editor/:id ─┐
                              │                      │
      ┌───────────────────────┼──────────────────────┼────────────────────┐
      ▼                       ▼                      ▼                    ▼
Catalog panel          3D viewport            Properties panel      Header actions
· Catalog              · top / perspective    · contextual by type  · Templates
· Collections          · transform gizmo      · AI Renders          · Save
· Table Designer       · radial menu          · Scene Lighting      · Undo / Redo
· AI Image to 3D       · perf overlay (admin) · builder panels      · Export ▸ PDF/PNG/Video
                                                                    · Share  ▸ link / iframe
Bottom toolbar — select · box · line · shapes · wall · lock ·        · AI Enhance
                 isolate · focus · grid · import ▸ · delete ▸

guest-entry ──▸ guest-editor          share/:token ──▸ read-only viewer / iframe embed
```

SECTION 07

Design system

Tokens, components and layout metrics, extracted from the compiled stylesheet.

7.1 Token architecture

Every colour is a CSS custom property prefixed `--qiso-`. The palette is **dark-first** (`--qiso-color-scheme: dark`), built on Tailwind's slate scale for surfaces and a single accent ramp exposed both as hex and as space-separated RGB triples so Tailwind opacity modifiers work.

SURFACES & INK

`--qiso-bg`	#020617
`--qiso-bg-soft`	#07111f
`--qiso-canvas-bg`	#050b18
`--qiso-surface`	rgba(15,23,42,.74)
`--qiso-surface-strong`	rgba(15,23,42,.92)
`--qiso-surface-muted`	rgba(30,41,59,.72)
`--qiso-border`	rgba(148,163,184,.16)
`--qiso-border-strong`	rgba(56,189,248,.36)
`--qiso-text`	#f8fafc
`--qiso-text-muted`	#94a3b8
`--qiso-text-subtle`	#64748b
`--qiso-grid`	rgba(255,255,255,.025)

ACCENT & SEMANTIC

`--qiso-primary`	#0ea5e9
`--qiso-primary-strong`	#0284c7
`--qiso-primary-soft`	rgba(14,165,233,.14)
`--qiso-primary-foreground`	#ffffff
`--qiso-success`	#10b981
`--qiso-warning`	#f59e0b
`--qiso-danger`	#ef4444
`--qiso-focus`	rgba(125,211,252,.75)
`--qiso-radius`	0.5rem
`--qiso-shadow`	0 24px 70px rgba(2,8,23,.36)

Each semantic colour also has `-soft` (12–14% fill) and `-border` (32–36%) variants — a tidy three-part pattern worth copying: *solid / soft fill / border*.

THE THEME IS SERVER-DRIVEN — VERIFIED

The stylesheet default is sky blue (`#0ea5e9`), but every live capture shows an **indigo/violet** accent. The bundle contains `GET /theme/config`, `GET /admin/theme`, `PATCH /admin/theme`, an `active_theme_id` field, and named presets including *"DesignCode Light"* and *"Qiso Blueprint"*. The admin console copy reads: "Choose the global Qiso palette. Users still control Dark or Light mode in Account."

So there are two independent layers: a globally administered palette, and a per-user light/dark preference. Nozila should build both; retrofitting runtime theming into hard-coded colours is miserable.

7.2 Component classes

All 57 `.qiso-*` *classes present in the compiled stylesheet, grouped by role.*

GROUP	CLASSES
Buttons	`qiso-button-primary` `qiso-button-secondary` `qiso-button-danger` `qiso-icon-button`
Form	`qiso-input` `qiso-select` `qiso-chip`
Containers	`qiso-panel` `qiso-panel-muted` `qiso-card` `qiso-card-hover` `qiso-page-shell` `qiso-page-header`

GROUP	CLASSES
Modal	`qiso-modal-overlay` `qiso-modal-panel` `qiso-modal-header` `qiso-modal-body` `qiso-modal-footer`
Table	`qiso-table-container` `qiso-table` `qiso-table-head-row` `qiso-table-row`
Notices	`qiso-notice` + `-neutral` `-success` `-warning` `-error`
Typography	`qiso-text-strong` `qiso-text-muted` `qiso-text-subtle`
Editor-specific	`qiso-editor-header` `qiso-editor-main` `qiso-editor-toolbar` `qiso-editor-panel` `qiso-editor-panel-header` `qiso-editor-panel-body` `qiso-editor-section` `qiso-editor-section-header` `qiso-editor-field` `qiso-editor-search` `qiso-editor-action` `qiso-editor-action-primary` `qiso-editor-icon-action` `qiso-editor-menu` `qiso-editor-menu-item` `qiso-editor-menu-label` `qiso-editor-segment` `qiso-editor-segment-button` `qiso-editor-segment-button-active` `qiso-editor-item-card` `qiso-editor-item-card-active` `qiso-editor-item-card-locked` `qiso-editor-thumb` `qiso-editor-muted-card` `qiso-editor-separator`

The editor has its own component vocabulary distinct from the rest of the app — denser padding, smaller type, tighter radii. That separation is deliberate and worth reproducing: dashboard components at editor density feel cramped, and editor components at dashboard density waste the panel.

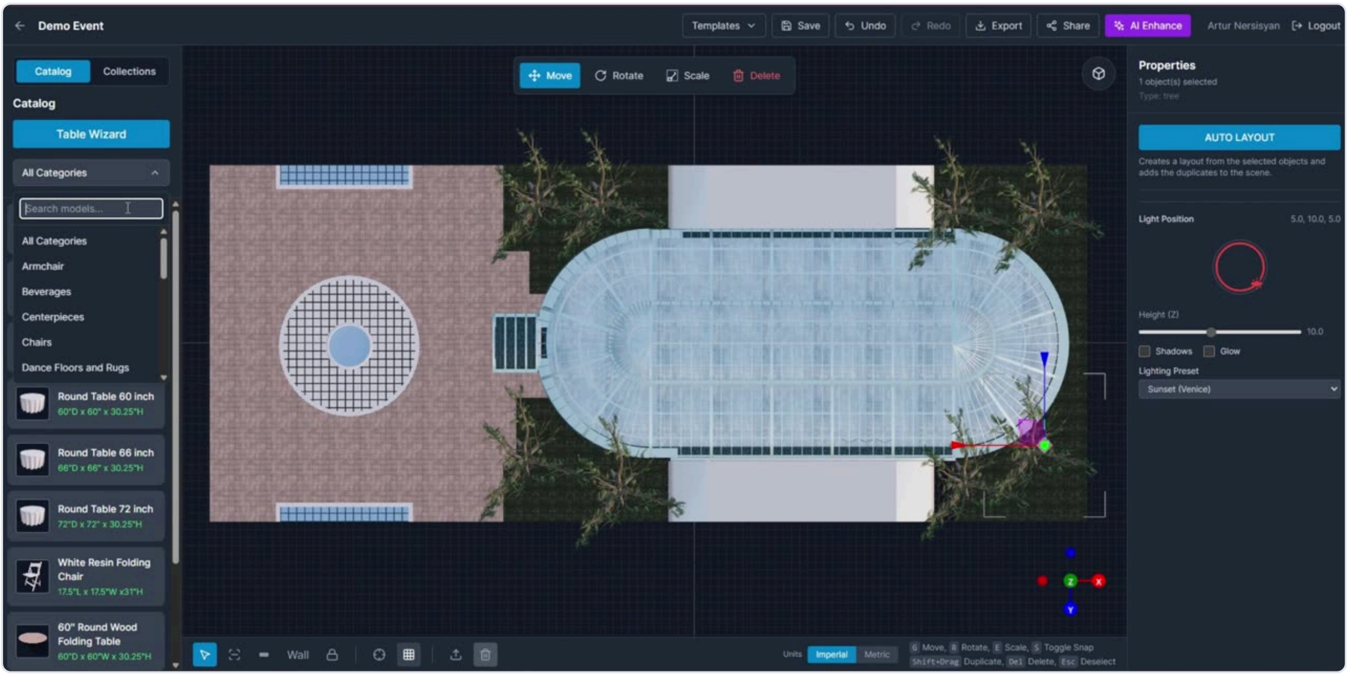
7.3 Typography and layout metrics

TYPE

- **Inter** throughout, weights 300–800 (marketing site loads exactly this range).
- Editor body text ≈ 12–13 px; labels ≈ 11 px; panel section headings ≈ 11 px uppercase with tracking.
- Canvas overlay labels use `18px Inter var`.

EDITOR METRICS (MEASURED FROM 1440-WIDE CAPTURES)

- Header height ≈ 48 px
- Left catalog panel ≈ 190–215 px
- Right properties panel ≈ 235–250 px
- Bottom toolbar ≈ 44 px, left-aligned icons
- Floating transform bar centred at ~24 px from top of canvas



The three-column editor shell. Left: catalog with category filter, search and item cards showing real dimensions. Centre: viewport with floating transform toolbar, view-cube control top-right, axis gizmo bottom-right. Right: contextual properties. Bottom: tool rail, unit toggle and a persistent keyboard legend. *(Earlier UI generation — note "Table Wizard" and "Auto Layout", since renamed.)*

SECTION 08

Domain model & MySQL schema

Entities derived from observed API paths and payload field names, expressed as MySQL 8 DDL.

Every field name in this section was observed in the production bundle — as a request-body key, a query parameter, a form-data field or a rendered value. Types, widths, nullability and indexes are engineering judgement; the *names and their existence* are evidence.

8.1 Entity map

```
company ──1:N──▸ user ──1:N──▸ project ──1:N──▸ plan ──1:1──▸ plan_scene (JSON)
   │                │                              │
   │                │                              ├──1:N──▸ plan_share
   │                │                              ├──0:1──▸ plan_floor_plan
   │                │                              ├──0:1──▸ plan_background
   │                │                              ├──0:1──▸ plan_venue          ──▸ venue
   │                │                              └──1:N──▸ ai_render            (AI Enhance history)
   │                │
   │                ├──1:N──▸ credit_ledger
   │                ├──1:N──▸ subscription
   │                ├──1:N──▸ ai_job          (all four AI job types)
   │                ├──1:N──▸ template
   │                └──1:N──▸ collection
   │
   ├──1:N──▸ company_member  (seat · role · visibility)
   ├──1:N──▸ company_invitation
   └──1:N──▸ company_setting (credit mode, asset flags)

catalog_category ──1:N──▸ catalog_item ──1:N──▸ catalog_texture_variation
texture_category ──1:N──▸ texture_asset
venue_generation_run ──1:N──▸ venue_revision_set ──▸ venue
```

8.2 Units and conventions

CONVENTION	RULE
Linear dimensions	**Integer millimetres**, suffix `_mm`. Observed: `width_mm` `height_mm` `depth_mm` `x_mm` `y_mm` `z_mm` `thickness_mm` `radius_mm` `diameter_mm` `length_mm` `offset_mm` `distance_mm` `deck_height_mm` `extrude_mm` `curve_depth_mm` `fold_amplitude_mm`
Angles	Degrees for UI (`angle_deg` , `sweep_deg`), radians in scene maths (`yaw_rad`). Rotation snap default **10°**.
World space	Metres for camera, lighting and origin (`origin_m` , `tile_size_m` , `scale_world_per_px`)
Model heights	Centimetres for generated-model transforms (`source_height_cm` , `target_height_cm`) — “internally stored in cm”
Display units	Per-user `preferred_units` : `metric` | `imperial` . Per-plan units also stored; changing the preference does not bulk-change saved plans.
Money	Integer minor units — `unit_amount` , `seat_unit_amount` (Stripe convention)
Timestamps	`created_at` `updated_at` `completed_at` `requested_at` `cycle_ends_at` `onboarding_completed_at` — UTC `DATETIME(3)`

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8.3 Core DDL

Identity, companies and access

```sql
CREATE TABLE company (
  id                  BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  name                VARCHAR(160) NOT NULL,
  main_email          VARCHAR(255),
  phone               VARCHAR(40),
  address_line        VARCHAR(255),
  city                VARCHAR(120),
  country             CHAR(2),                      -- 2-letter code, validated client-side
  postal_code         VARCHAR(20),
  logo_url            VARCHAR(512),                 -- PNG/JPG/WEBP, max 10 MB, resized to 512 px
  credit_mode         ENUM('per_user','shared_pool') NOT NULL DEFAULT 'per_user',
  pending_credit_mode ENUM('per_user','shared_pool') NULL,  -- applies next cycle
  pool_balance        INT NOT NULL DEFAULT 0,
  global_models_enabled            TINYINT(1) NOT NULL DEFAULT 1,
  global_textures_enabled          TINYINT(1) NOT NULL DEFAULT 1,
  company_admin_can_manage_assets  TINYINT(1) NOT NULL DEFAULT 0,
  ai_image_to_3d_enabled           TINYINT(1) NOT NULL DEFAULT 1,
  venue_generation_access_override ENUM('default','granted','blocked') NOT NULL DEFAULT 'default',
  stripe_customer_id VARCHAR(64),
  is_active           TINYINT(1) NOT NULL DEFAULT 1,
  created_at          DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  updated_at          DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3) ON UPDATE CURRENT_TIMESTAMP(3)
) ENGINE=InnoDB;

CREATE TABLE user (
  id                  BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  company_id          BIGINT UNSIGNED NULL,
  email               VARCHAR(255) NOT NULL UNIQUE,
  password_hash       VARCHAR(255) NULL,          -- null for Google-only accounts
  google_sub          VARCHAR(64) NULL UNIQUE,
  first_name          VARCHAR(60) NOT NULL,
  last_name           VARCHAR(60) NOT NULL,        -- combined max 100 chars, enforced
  display_name        VARCHAR(120),                -- min 2 chars
  phone               VARCHAR(40),
  role                ENUM('user','company_admin','super_admin') NOT NULL DEFAULT 'user',
  plan_tier           ENUM('free','plus','pro') NOT NULL DEFAULT 'free',
  preferred_units     ENUM('metric','imperial') NOT NULL DEFAULT 'imperial',
  appearance          ENUM('dark','light') NOT NULL DEFAULT 'dark',
  is_email_verified TINYINT(1) NOT NULL DEFAULT 0,
  is_blocked          TINYINT(1) NOT NULL DEFAULT 0,
  marketing_opt_in    TINYINT(1) NOT NULL DEFAULT 0,
  onboarding_role_id        BIGINT UNSIGNED NULL,    -- "I am a"
  onboarding_design_ids     JSON NULL,               -- "I mainly design" (multi-select)
  onboarding_completed_at DATETIME(3) NULL,
  credits_remaining         INT NOT NULL DEFAULT 0,  -- CACHE of credit_ledger; never authoritative
  monthly_credit_quota      INT NOT NULL DEFAULT 0,  -- may be manually overridden by admin
  created_at          DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  updated_at          DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3) ON UPDATE CURRENT_TIMESTAMP(3),
  KEY ix_user_company (company_id),
  KEY ix_user_plan (plan_tier, is_blocked),
  CONSTRAINT fk_user_company FOREIGN KEY (company_id) REFERENCES company(id) ON DELETE SET NULL
) ENGINE=InnoDB;

CREATE TABLE company_member (
```

```sql
  id             BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  company_id     BIGINT UNSIGNED NOT NULL,
  user_id        BIGINT UNSIGNED NULL,          -- null while invitation pending
  invited_email VARCHAR(255) NOT NULL,
  role           ENUM('member','admin') NOT NULL DEFAULT 'member',
  visibility     ENUM('own_only','company_wide') NOT NULL DEFAULT 'own_only',
  seat_plan_tier ENUM('free','plus','pro') NOT NULL,
  status         ENUM('invited','active','pending_payment','payment_failed','cancelled') NOT NULL,
  invitation_token    CHAR(43) NULL UNIQUE,
  invitation_expires_at DATETIME(3) NULL,
  stripe_subscription_item_id VARCHAR(64),
  created_at     DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  UNIQUE KEY uq_member (company_id, invited_email),
  KEY ix_member_user (user_id)
) ENGINE=InnoDB;
```

Projects, plans and the scene document

```sql
CREATE TABLE project (
  id           BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  owner_id     BIGINT UNSIGNED NOT NULL,
  company_id   BIGINT UNSIGNED NULL,
  title        VARCHAR(180) NOT NULL,
  description TEXT NULL,
  created_at   DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  updated_at   DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3) ON UPDATE CURRENT_TIMESTAMP(3),
  KEY ix_project_owner (owner_id, updated_at DESC)
) ENGINE=InnoDB;
-- Deletion rule observed in UI: a project with plans cannot be deleted ("Delete all plans first").

CREATE TABLE plan (
  id             BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  project_id     BIGINT UNSIGNED NOT NULL,
  owner_id       BIGINT UNSIGNED NOT NULL,
  title          VARCHAR(180) NOT NULL,
  editor_type    ENUM('3d','2d') NOT NULL DEFAULT '3d',    -- UI label: "Sketch Type"
  units          ENUM('metric','imperial') NOT NULL,           -- copied from user pref at creation
  start_in_top_view TINYINT(1) NOT NULL DEFAULT 0,
  preview_url    VARCHAR(512) NULL,                          -- dashboard thumbnail
  is_read_only   TINYINT(1) NOT NULL DEFAULT 0,
  object_count   INT NOT NULL DEFAULT 0,
  created_at     DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  updated_at     DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3) ON UPDATE CURRENT_TIMESTAMP(3),
  KEY ix_plan_project (project_id, updated_at DESC),
  CONSTRAINT fk_plan_project FOREIGN KEY (project_id) REFERENCES project(id) ON DELETE CASCADE
) ENGINE=InnoDB;

CREATE TABLE plan_scene (
  plan_id        BIGINT UNSIGNED PRIMARY KEY,
  schema_version SMALLINT UNSIGNED NOT NULL DEFAULT 1,
  scene          JSON NOT NULL,      -- objects, groups, transforms, materials
  wall_blueprint JSON NULL,         -- spans, segments, openings, floor polygons
  lighting       JSON NULL,         -- preset, position, height, shadows, glow
  camera         JSON NULL,         -- mode, position, target, saved camera path
  updated_at     DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3) ON UPDATE CURRENT_TIMESTAMP(3),
  CONSTRAINT fk_scene_plan FOREIGN KEY (plan_id) REFERENCES plan(id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

SCENE OBJECT SHAPE

Each entry in `scene.objects[]` carries, at minimum, the fields observed in the editor's properties panel and store:

```
{
  "id": "uuid",
  "type": "catalog | shape | wall | floor | curtain | tent | biljax-stage | text | opening",
  "catalog_item_id": 1042,
  "group_id": "group-table-group-7",        // Table Designer keeps sets together
  "position_mm": { "x": 0, "y": 0, "z": 0 },
  "rotation_deg": { "x": 0, "y": 90, "z": 0 },
  "scale": { "x": 1, "y": 1, "z": 1 },
  "dimensions_mm": { "width": 1524, "depth": 1524, "height": 768 },
  "locked": false,
  "opacity": 1.0,
  "material_color_config": { "<material_id>": "#c9b08a" },
  "texture_variation_id": 88,
  "texture_tiling_enabled": true,
  "texture_tile_size_m": 0.5,
  "params": { /* type-specific: curtain, stage, tent, shape — see §12 */ }
}
```

Catalogue and textures

```sql
CREATE TABLE catalog_category (
  id         BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  name       VARCHAR(120) NOT NULL,
  slug       VARCHAR(140) NOT NULL UNIQUE,
  sort_order INT NOT NULL DEFAULT 0
) ENGINE=InnoDB;

CREATE TABLE catalog_item (
  id            BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  category_id   BIGINT UNSIGNED NOT NULL,
  company_id    BIGINT UNSIGNED NULL,     -- null = global library
  owner_id      BIGINT UNSIGNED NULL,     -- set for personal "My catalog" items
  scope         ENUM('global','company','personal') NOT NULL,
  name          VARCHAR(180) NOT NULL,
  description   TEXT,
  model_url     VARCHAR(512) NOT NULL,   -- GLB/GLTF/OBJ/FBX, max 50 MB on upload
  preview_image VARCHAR(512),               -- JPG/PNG/WebP, max 5 MB
  icon_url      VARCHAR(512),
  width_mm      INT, depth_mm INT, height_mm INT, diameter_mm INT,
  table_shape   ENUM('round','rectangular','other') NULL,
  seats_default SMALLINT NULL,
  is_free_plan_available TINYINT(1) NOT NULL DEFAULT 0,   -- "Free users can place this item"
  default_material_id VARCHAR(120) NULL,
  material_color_config JSON NULL,
  variations_manifest   JSON NULL,
  texture_variation_count SMALLINT NOT NULL DEFAULT 0,
  is_active     TINYINT(1) NOT NULL DEFAULT 1,
  review_status ENUM('pending','approved','rejected') NOT NULL DEFAULT 'approved',
  created_at    DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  KEY ix_item_cat (category_id, is_active),
  KEY ix_item_scope (scope, company_id),
  FULLTEXT KEY ft_item (name, description)
) ENGINE=InnoDB;

CREATE TABLE catalog_texture_variation (
  id              BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  catalog_item_id BIGINT UNSIGNED NOT NULL,
  name            VARCHAR(140) NOT NULL,   -- every variation must be named
  material_id     VARCHAR(120) NOT NULL,
  texture_url     VARCHAR(512) NOT NULL,
  is_default      TINYINT(1) NOT NULL DEFAULT 0,
  KEY ix_var_item (catalog_item_id)
) ENGINE=InnoDB;

CREATE TABLE texture_asset (
  id          BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  category_id BIGINT UNSIGNED NOT NULL,
  company_id  BIGINT UNSIGNED NULL,       -- null = global library
  name        VARCHAR(160) NOT NULL,
  description VARCHAR(400) NOT NULL,      -- "Each texture needs a description"
  image_url   VARCHAR(512) NOT NULL,      -- PNG/JPEG/WebP
  pbr_enabled TINYINT(1) NOT NULL DEFAULT 0,
  tile_size_m DECIMAL(6,3) NULL,
  KEY ix_tex_cat (category_id, company_id)
) ENGINE=InnoDB;
```

Credits, subscriptions and the AI job queue

```
CREATE TABLE credit_ledger (
  id          BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  user_id     BIGINT UNSIGNED NOT NULL,
  company_id  BIGINT UNSIGNED NULL,          -- set when drawn from / added to the pool
  delta       INT NOT NULL,                   -- +grant/+purchase/+refund, -debit
  reason      ENUM('cycle_grant','purchase','debit','refund',
                   'transfer_in','transfer_out','admin_adjustment') NOT NULL,
  feature_code VARCHAR(48) NULL,              -- ai_image_to_3d | ai_enhance | ...
  ai_job_id   BIGINT UNSIGNED NULL,
  expires_at  DATETIME(3) NULL,               -- purchased packs can expire ("One month expiry")
  balance_after INT NOT NULL,
  created_at  DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  KEY ix_ledger_user (user_id, created_at DESC),
  KEY ix_ledger_feature (feature_code, created_at)
) ENGINE=InnoDB;

CREATE TABLE credit_ratio (             -- admin-editable cost per action
  feature_code VARCHAR(48) PRIMARY KEY,
  label        VARCHAR(120) NOT NULL,
  description  VARCHAR(400),
  credits      INT NOT NULL,
  is_active    TINYINT(1) NOT NULL DEFAULT 1
) ENGINE=InnoDB;
-- Seed: ai_image_to_3d=10, floor_plan_ai_draw=3, ai_enhance=2, google_maps=1,
--       ai_enhance_nano_banana_2, ai_enhance_qwen, venue_generation

CREATE TABLE credit_pack (              -- admin-editable purchasable packs
  id             BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  credit_amount  INT NOT NULL,                -- e.g. 50 / 100 / 150
  unit_amount    INT NOT NULL,                -- USD cents
  stripe_price_id VARCHAR(64) NOT NULL,
  historical_price_ids JSON NULL,
  expiry_days    INT NULL,
  is_active      TINYINT(1) NOT NULL DEFAULT 1,
  display_order  INT NOT NULL DEFAULT 0
) ENGINE=InnoDB;

CREATE TABLE subscription (
  id            BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  user_id       BIGINT UNSIGNED NULL,
  company_id    BIGINT UNSIGNED NULL,
  plan_tier     ENUM('free','plus','pro') NOT NULL,
  billing_source ENUM('stripe','manual','company') NOT NULL DEFAULT 'stripe',
  stripe_subscription_id     VARCHAR(64),
  stripe_subscription_status VARCHAR(32),
  billing_cycle_anchor DATETIME(3),
  cycle_ends_at        DATETIME(3),
  monthly_credit_quota INT NOT NULL DEFAULT 0,
  cancel_at_period_end TINYINT(1) NOT NULL DEFAULT 0,
  cancellation_reason_id BIGINT UNSIGNED NULL,
  cancellation_note    TEXT NULL,
  created_at    DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3)
) ENGINE=InnoDB;

CREATE TABLE ai_job (                    -- ONE table for all four AI features
  id            BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
```

```sql
  public_id     CHAR(36) NOT NULL UNIQUE,     -- job_id / run_id exposed to client
  user_id       BIGINT UNSIGNED NOT NULL,
  company_id    BIGINT UNSIGNED NULL,
  plan_id       BIGINT UNSIGNED NULL,
  project_id    BIGINT UNSIGNED NULL,
  process_type  ENUM('ai_image_to_3d','ai_enhance','floor_plan_ai_draw',
                     'venue_generation','google_maps') NOT NULL,
  provider      VARCHAR(64),                   -- world_labs | planner_ai | ...
  provider_model VARCHAR(64),                  -- marble-1.0-draft | nano_banana_2 | qwen
  status        ENUM('queued','in_progress','completed','failed','cancelled') NOT NULL,
  progress      TINYINT UNSIGNED NOT NULL DEFAULT 0,
  idempotency_key VARCHAR(80) NOT NULL,
  credits_charged INT NOT NULL DEFAULT 0,
  input_payload  JSON,                         -- prompt, preset_ids, input_mode, source refs
  output_payload JSON,                         -- artefact urls, mesh meta, wall segments
  source_image_filename VARCHAR(255),
  error_code    VARCHAR(64),
  error_message TEXT,
  trace_id      CHAR(32),
  created_at    DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  completed_at  DATETIME(3) NULL,
  UNIQUE KEY uq_idem (user_id, idempotency_key),
  KEY ix_job_type (process_type, status, created_at DESC),
  KEY ix_job_user (user_id, created_at DESC)
) ENGINE=InnoDB;
```

Sharing, templates, collections and venues

```
CREATE TABLE plan_share (
  id         BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  plan_id    BIGINT UNSIGNED NOT NULL,
  token      CHAR(43) NOT NULL UNIQUE,   -- opaque, NOT the plan id
  mode       ENUM('view','embed') NOT NULL DEFAULT 'view',
  expires_at DATETIME(3) NULL,
  revoked_at DATETIME(3) NULL,
  created_at DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3)
) ENGINE=InnoDB;

CREATE TABLE template (
  id          BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  owner_id    BIGINT UNSIGNED NULL,
  scope       ENUM('local','global') NOT NULL DEFAULT 'local',
  title       VARCHAR(180) NOT NULL,   -- min 3 chars; unique per scope
  description TEXT,
  snapshot    JSON NOT NULL,           -- full plan scene snapshot
  preview_url VARCHAR(512),
  created_at  DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  UNIQUE KEY uq_tpl (scope, owner_id, title)   -- error TEMPLATE_TITLE_EXISTS
) ENGINE=InnoDB;

CREATE TABLE collection (
  id            BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  owner_id      BIGINT UNSIGNED NOT NULL,
  name          VARCHAR(180) NOT NULL,
  objects       JSON NOT NULL,         -- relative placement preserved
  object_count  INT NOT NULL,
  preview_url   VARCHAR(512),               -- captured from the 3D canvas
  source_plan_id BIGINT UNSIGNED NULL,
  created_at    DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3)
) ENGINE=InnoDB;

CREATE TABLE venue (                   -- reusable AI-generated environment
  id          BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
  owner_id    BIGINT UNSIGNED NULL,
  company_id  BIGINT UNSIGNED NULL,
  scope       ENUM('public','company','personal') NOT NULL,
  name        VARCHAR(180) NOT NULL,   -- default "Untitled venue"
  splat_url   VARCHAR(512) NOT NULL,   -- SPZ Gaussian splat
  preview_url VARCHAR(512),
  scale_profile_id BIGINT UNSIGNED NULL,
  source_run_id BIGINT UNSIGNED NULL,
  created_at  DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3)
) ENGINE=InnoDB;

CREATE TABLE plan_venue (              -- attachment + per-plan transform
  plan_id   BIGINT UNSIGNED PRIMARY KEY,
  venue_id  BIGINT UNSIGNED NOT NULL,
  transform JSON NOT NULL,           -- position, rotation, scale
  is_locked TINYINT(1) NOT NULL DEFAULT 1,
  is_hidden TINYINT(1) NOT NULL DEFAULT 0
) ENGINE=InnoDB;
```

8.4 Supporting tables

TABLE	PURPOSE AND KEY COLUMNS
`plan_floor_plan`	Calibrated drawing: `image_url`, `file_key`, `scale_world_per_px`, `real_distance`, `point_a` / `point_b`, `origin_m`, locked flag
`plan_background`	Un-scaled backdrop image: `image_url`, `file_key`. One per plan; replacing deletes the old file.
`ai_render`	AI Enhance history per plan: `job_id`, `prompt`, `model`, `preview_url`, `media_url`, `created_at`
`onboarding_option`	`section` (1|2), `label`, `sort_order`, `is_active` — the "I am a" / "I mainly design" choices
`onboarding_template_assignment`	`section_1_option_id` × `section_2_signature` → `template_id`; the grid that picks a starter workspace
`cancellation_reason`	`label`, `requires_note`, `sort_order`, `is_active`
`subscription_request`	Manual upgrade queue: `user_id`, `requested_plan`, `status`, `requested_at`, `decided_by`
`theme_preset`	`name`, `tokens` JSON, `is_active` — e.g. "DesignCode Light", "Qiso Blueprint"
`upload`	`file_key`, `owner_id`, `content_type`, `size`, `purpose`. Deleted explicitly via `DELETE /uploads/{file_key}`.
`analytics_event`	`anonymous_id`, `user_id`, `name`, `props` JSON — see Appendix B for the observed event vocabulary
`guest_session`	`session_id`, `name`, `email`, `background_image`, `camera` JSON (extracted perspective). Expires; cleaned up periodically.

STORAGE SIZING NOTE

Venue splats are the dominant storage cost — the subject ships a 5 MB renderer, and SPZ assets themselves typically run tens of megabytes. Keep every binary artefact (GLB, textures, renders, exports, splats) in object storage and hold only URLs and keys in MySQL. Never put binaries in the database.

SECTION 09

REST API surface

Every endpoint observed in the production bundle, grouped by domain. Paths are verbatim.

This is the single most valuable artefact in this document. Roughly 180 endpoints were recovered by reading axios call sites in the shipped JavaScript. Paths and request payload keys are **observed fact**; response shapes are inferred from how the client consumes them and are marked where uncertain.

9.1 Authentication & account

METHOD	PATH	NOTES
POST	`/auth/register`	Individual or Company. Company variant adds `companyName`, `companyAddress`, `companyCountry`, `companyPostalCode`, `phone`, main email. reCAPTCHA token required.
POST	`/auth/login`	Email + password; reCAPTCHA
POST	`/auth/google`	Google credential exchange
GET	`/auth/me`	Session user: profile, role, plan, credits, company, flags
PATCH	`/auth/me`	`display_name`, `preferred_units`, appearance
POST	`/auth/me/password`	Requires current password; new ≥ 8 chars
POST	`/auth/forgot`	Send reset link
POST	`/auth/reset`	Token + new password
POST	`/auth/verify`	Email verification token
POST	`/auth/verification/resend`	Rate-limited
GET	`/auth/onboarding/section-1`	“I am a” options
GET	`/auth/onboarding/section-2`	“I mainly design” options
POST	`/auth/onboarding/bootstrap-assigned-template`	Creates the starter project + “Wellcome plan” from the answer pair
GET	`/privacy/consent`	Consent capture
POST	`/privacy/consent`	

9.2 Projects & plans

METHOD	PATH	NOTES
GET	`/projects/`	Project list with recent plan previews
POST	`/projects/`	`title`, optional `description`
PATCH	`/projects/{id}`	Rename / re-describe
DEL	`/projects/{id}`	Blocked while plans exist
GET	`/plans/?project_id={id}`	Plans in a project
GET	`/plans/`	All plans for the user
POST	`/plans/`	`title`, `project_id`, `editor_type`
PATCH	`/plans/{id}`	**The save endpoint** — scene document, walls, lighting, camera
DEL	`/plans/{id}`	—

METHOD	PATH	NOTES
GET	`/plans/share/{token}`	Public read-only fetch
POST	`/plans/{id}/share`	Mint a share token; blocked on read-only plans
POST	`/plans/{id}/export`	PDF / PNG / video; returns a download URL

9.3 Catalogue, textures, templates & collections

METHOD	PATH	NOTES
GET	`/catalog/categories`	Category list / create
POST	`/catalog/categories`	
GET	`/catalog/items?…`	Params: `q`, `category_id`, `category_slug`, `scope`, `company_id`, `limit`, `offset`, `include_texture_variations`, `tables-with-linens` mode
GET	`/catalog/items/by-ids?…`	Bulk hydrate on scene load
GET	`/catalog/items/{id}/model`	Signed model URL (refreshed on expiry)
GET	`/catalog/items/{id}/texture-variations`	—
POST	`/catalog/items/upload-with-model`	multipart: model (OBJ/FBX/GLB/GLTF, ≤ 50 MB), preview (≤ 5 MB), `texture_files`, metadata
PUT	`/catalog/items/{id}/update-with-model`	Replace model and/or variations
DEL	`/catalog/items/{id}`	—
GET	`/textures/categories`	Texture library
GET	`/textures/assets`	
POST	`/textures/sets`	
PATCH	`/textures/assets/{id}`	
DEL	`/textures/assets/{id}`	—
GET	`/templates/`	Local + global
POST	`/templates/`	Save current plan as template
PATCH	`/templates/{id}`	Overwrite / rename
DEL	`/templates/{id}`	—
POST	`/templates/{id}/apply-to-plan`	Replaces current scene objects
GET	`/collections/`	Object-group collections
POST	`/collections/`	
GET	`/collections/{id}`	
DEL	`/collections/{id}`	—
GET	`/articles/by-key/{key}`	Contextual in-app help lookup
GET	`/theme/config`	Active palette tokens

9.4 Uploads & site context

METHOD	PATH	NOTES
POST	`/uploads/background`	PNG/JPG/WebP/GIF/SVG, ≤ 10 MB. Returns `file_key` .
POST	`/uploads/google-maps-static`	Server-side Static Maps capture; returns image + scale metadata
DEL	`/uploads/{file_key}`	Explicit cleanup on replace/remove
POST	`/plans/{id}/floor-plan/ai-draw`	Start wall tracing. Raster only — SVG rejected.
GET	`/plans/{id}/floor-plan/ai-draw/{jobId}`	Poll status → wall segments

9.5 AI: Enhance, Image-to-3D, Venue Generation

METHOD	PATH	NOTES
POST	`/plans/{id}/ai-enhance`	Body: canvas image, `prompt` , `model`
GET	`/plans/{id}/ai-enhance/{jobId}`	Poll
GET	`/plans/{id}/ai-enhance/history`	Powers the "AI Renders" panel
POST	`/plans/guest/{sessionId}/ai-enhance`	Guest-mode variant
POST	`/model-generation/upload-image-v3`	multipart: image (JPG/PNG/WebP ≤ 10 MB), `pbr_enabled` , model tier. Returns `job_id` .
GET	`/model-generation/status/{job_id}`	Poll
GET	`/model-generation/history`	Past runs, reopenable
GET	`/model-generation/history/{id}`	
POST	`/model-generation/save-to-catalog`	`name` , `description` , `category_id` , `scope` , `company_id` , `target_height_cm` , `anchor_mode` , `transformed_glb` , `preview_image`
GET	`/projects/{id}/venue-generation-capabilities`	Access + credit preflight
POST	`/projects/{id}/venue-generation-runs`	multipart. `input_mode` : `single` | `multi_view` | `panorama` . Idempotency-Key header.
GET	`/projects/{id}/venue-generation-runs?limit=20`	Run history
GET	`/venue-generation-runs/{id}`	Poll a run
POST	`/venue-generation-runs/{id}/suggestions`	Body: `revision_set_id` , `idempotency_key`
POST	`/venue-generation-runs/{id}/suggestions/skip`	—
POST	`/venue-generation-runs/{id}/image-revisions`	Body: `parent_revision_set_id` , `prompt` , `preset_ids` . One cleanup included per run.
POST	`/venue-generation-runs/{id}/generate`	Body: `revision_set_id` , `model` , `name`
POST	`/venue-generation-runs/{id}/cancel`	Refunds credits
POST	`/venue-generation-runs/{id}/finish`	Body: `name` , `scale_profile_id`
GET	`/projects/{id}/venues`	Reusable venue library

METHOD	PATH	NOTES
PUT	`/plans/{id}/venue`	Attach. Body: `venue_id`, `replace`
GET	`/plans/{id}/venue`	Accepts `share_token` for public views
PATCH	`/plans/{id}/venue/transform`	Position / rotation / scale of the environment
DEL	`/plans/{id}/venue`	Detach; venue stays in the library

9.6 Subscription, credits & company billing

METHOD	PATH	NOTES
GET	`/subscription/pricing`	Live tier prices; checkout disabled if unavailable
GET	`/subscription/credit-ratios`	Cost per action, for display
GET	`/subscription/cancellation-reasons`	Required reason list
POST	`/subscription/checkout-session`	Plan checkout
POST	`/subscription/checkout-sessions/{id}/complete`	
POST	`/subscription/checkout-sessions/{id}/complete-onboarding`	Onboarding-specific return path
POST	`/subscription/credit-checkout-session`	Credit-pack purchase
POST	`/subscription/credit-checkout-sessions/{id}/complete`	
POST	`/subscription/customer-portal`	Stripe portal redirect
POST	`/subscription/cancel`	Reason + optional note; schedules at period end
GET	`/company/profile`	Company details
PATCH	`/company/profile`	
POST	`/company/logo`	≤ 10 MB, resized to 512 px, used on PDF exports and embeds
DEL	`/company/logo`	
POST	`/company/convert`	Upgrade an individual account to a company workspace
GET	`/company/members`	Team list
PATCH	`/company/members/{id}`	Role / visibility
DEL	`/company/members/{id}`	Projects transfer to the company owner
POST	`/company/seats`	Add a seat; company card charged before invite is sent
POST	`/company/seats/preview`	
POST	`/company/seats/checkout-sessions/{id}/complete`	—
POST	`/company/members/{id}/upgrade`	Move a member Plus → Pro mid-cycle, with proration preview
POST	`/company/members/{id}/upgrade/preview`	
POST	`/company/members/{id}/credits/checkout-session`	Buy credits for one member
POST	`/company/member-credit-checkout-sessions/{id}/complete`	
PATCH	`/company/settings/credit-mode`	Per-user ⇄ shared pool; applies next cycle

METHOD	PATH	NOTES
GET	`/company/credits/history`	Usage by member and feature; CSV export
GET	`/company/credits/history/export`	
GET	`/company/billing/expenses`	Company spend
POST	`/company/billing/customer-portal`	Stripe portal for the company
GET	`/company/invitations/preview`	Invitation landing + acceptance
POST	`/company/invitations/accept`	

9.7 Administration

All 26 admin endpoint families, recovered from the Super Admin Console chunk and the main bundle.

METHOD	PATH	CONSOLE SECTION
GET	`/admin/analytics/dashboard`	Analytics
GET	`/admin/analytics/registered-users`	
GET	`/admin/analytics/registered-users/{id}/events`	
GET	`/admin/users?…`	Users — search, filter by company / role / status, paginate
POST	`/admin/users`	
PATCH	`/admin/users/{id}`	
DEL	`/admin/users/{id}`	
POST	`/admin/users/{id}/impersonate`	
POST	`/admin/users/{id}/reset-password`	
GET	`/admin/companies`	Companies + tenant asset flags
POST	`/admin/companies`	
PATCH	`/admin/companies/{id}`	
DEL	`/admin/companies/{id}`	
GET	`/admin/companies/{id}/users?sort=created_desc`	Company Users
GET	`/admin/company/users?…`	Company Users drill-down
GET	`/admin/company/users/{id}/projects`	
GET	`/admin/company/users/{id}/plans`	
GET	`/admin/company/projects/{id}/plans`	—
GET	`/admin/catalog-reviews?…`	Model & Venue Review
GET	`/admin/catalog-reviews/{id}`	
POST	`/admin/catalog-reviews/{id}/approve`	
POST	`/admin/catalog-reviews/{id}/reject`	
GET	`/admin/subscription-billing-prices`	Stripe Pricing — creates new recurring Prices
PATCH	`/admin/subscription-billing-prices`	
GET	`/admin/subscription-plan-credits`	Plan monthly credit quotas
PATCH	`/admin/subscription-plan-credits`	

METHOD	PATH	CONSOLE SECTION
GET	`/admin/subscription-credit-packs`	Purchasable packs
PATCH	`/admin/subscription-credit-packs`	
GET	`/admin/subscription-credit-ratios`	**Cost per AI action**
PATCH	`/admin/subscription-credit-ratios`	
GET	`/admin/subscription-users`	Paid Subscribers
GET	`/admin/subscription-requests`	Manual upgrade approvals
POST	`/admin/subscription-requests/{id}/approve`	
POST	`/admin/subscription-requests/{id}/reject`	
GET	`/admin/subscription-cancellations`	Cancellation Feedback
GET	`/admin/subscription-cancellation-reasons`	Reason catalogue (`requires_note` , order)
POST	`/admin/subscription-cancellation-reasons`	
PATCH	`/admin/subscription-cancellation-reasons/{id}`	
GET	`/admin/onboarding/section-1`	Onboarding options + starter-template grid
POST	`/admin/onboarding/section-1`	
PATCH	`/admin/onboarding/section-1/{id}`	
DEL	`/admin/onboarding/section-1/{id}`	
GET	`/admin/onboarding/template-assignments/grid`	
PUT	`/admin/onboarding/template-assignments`	
DEL	`/admin/onboarding/template-assignments/{id}`	*(section-2 mirrors section-1 exactly)*
GET	`/admin/reports/model-generations?…`	Operational AI reporting, filterable and searchable
GET	`/admin/reports/venue-generations?…`	
GET	`/admin/reports/ai-process-runs?…`	
GET	`/admin/theme`	Global palette preset
PATCH	`/admin/theme`	

9.8 Cross-cutting conventions

CONCERN	CONVENTION
Pagination	`?limit=&offset=` ; responses carry `has_more` . Infinite scroll in catalogue panels ("Scroll to load more…").
Sorting	Token strings — `created_desc`
Filtering	`created_from` / `created_to` , `status` , `company_id` , `mode` , free-text `q`

CONCERN	CONVENTION
Idempotency	`Idempotency-Key` header on multipart AI posts; `idempotency_key` body field on JSON AI posts. Client-generated as `{action}-{uuid}` .
Error codes	`PLAN_UPGRADE_REQUIRED` · `PLAN_REQUEST_PENDING` · `INSUFFICIENT_CREDITS` · `TEMPLATE_TITLE_EXISTS` · `provider_payment_required` · `payment_failed`
Access failures	403 and 404 are treated identically by the client's retry logic — retries are suppressed, and the user is redirected to `/dashboard`
Public access	Share endpoints accept a `share_token` query parameter instead of a bearer token

SECTION 10

The editor

Shell, viewport, camera, selection, transforms and history — the 80% of the product.

10.1 Shell anatomy

```
┌──────────────────────────────────────────────────────────────────────────────────┐
│ ←  Plan title          Templates▾ Save Undo Redo Export▾ Share▾ ✦AI Enhance  ● user▾│  header
├──────────────┬──────────────────────────────────────────────┬────────────────────┤
│ Table        │        ⟦ Move · Rotate · Scale · Delete ⟧     │ Properties         │
│ Designer     │                                        ⊞     │  N selected        │
│              │                                              │ Type:          │
│ Catalog  ✦AI │                                              │ ┌────────────────┐ │
│ ┌Catalog│Coll│                 3D VIEWPORT                  │ │ AI Renders (n) │ │
│ │category ▾ │                                               │ ├────────────────┤ │
│ │🔍  search │                                               │ │ contextual     │ │
│ ├───────────┤                                               │ │ builder panel  │ │
│ │ ▢ item    │                                               │ ├────────────────┤ │
│ │ ▢ item    │                                               │ │ Quick Layout   │ │
│ │ ▢ item …  │                                        ⊕XYZ   │ ├────────────────┤ │
│ └───────────┘                                               │ │ Scene Lighting │ │
│              │                                              │ └────────────────┘ │
├──────────────┴──────────────────────────────────────────────┴────────────────────┤
│ ↖ ⬚ — △ Wall 🔒 ◎ ⊞ 🠕 🗑       Units[Imperial|Metric]   G Move R Rotate E Scale … │  toolbar
└──────────────────────────────────────────────────────────────────────────────────┘
```

Header (left → right)

- **Back arrow** → “Back to project”
- **Plan title**, inline editable
- **Templates ▾** — Save Template · Load Template · Save as Collection · Edit/Delete Template. Split by scope: *My Templates* (local, private) and *Global Templates* (available to every user).
- **Save** — with states *Saving… / Saved / Save failed / Retry save*. Autosave runs alongside (*Autosaved / Autosave failed*).
- **Undo · Redo** — disabled when the stack is empty
- **Export ▾** — PDF · PNG · Video
- **Share ▾** — Copy share link · Embed iframe
- **AI Enhance** — visually distinct (gradient), with *Enhancing…* state
- **User menu** — avatar initials, tier badge (Free/Plus/Pro/Super Admin), name

Left panel

- **Table Designer** — full-width primary button, always on top
- **“Catalog”** heading with an **AI Image to 3D** button beside it
- Tabs: **Catalog** | **Collections**
- Category `<select>` (“All Categories”, Tents, Stages, Chairs, Beverages, Centerpieces, Dance Floors and Rugs, Decor & Accessories, Armchair, Linens…)
- Search field with live typeahead suggestions above the result list
- Item cards: thumbnail, name, and a green sub-line carrying **real dimensions** (e.g. `60"D × 60" × 30.25"H`) or a descriptor and variation count (`4× (4' × 4') Panels · 3 Variations`)
- Infinite scroll — “Scroll to load more”, “Loading more items…”
- Locked items render in a distinct locked state: “This catalog model is available on Plus and Pro plans.”
- **Admin mode** adds an “Upload Model” button and per-card pencil/trash affordances

Right panel — a stack of contextual cards

1. **AI Renders (n)** — collapsible; "Review previous AI Enhance results for this plan." Empty: "No AI renders yet."
2. **Selection** — "N object(s) selected · Type: *slug*", or the empty state "Select an object on the canvas to edit transforms, dimensions, and materials."
3. **Contextual editor** — the builder panel for the selected type (§11–12)
4. **Quick Layout** — "Creates a Quick Layout arrangement from the selected objects and adds the duplicates to the scene."
5. **Scene Lighting** — "Global controls for the full plan."

Bottom toolbar

TOOL	ANALYTICS ID	BEHAVIOUR
Select	`editor.toolbar.select`	Default; single click selects
Selection Box	`editor.toolbar.select-box`	Drag-rectangle multi-select, best in top view
Line Placement	`editor.toolbar.line-placement`	Repeat an item along a line
Shapes	`editor.toolbar.shapes`	Disabled while wall editing is open
Wall	`editor.toolbar.wall-draw`	3D only; opens the Walls panel
Lock / Unlock	`editor.toolbar.lock-toggle`	Prevents accidental movement
Isolate	`editor.toolbar.isolate`	Hides everything else — *including floor plan and background*
Focus Selected	`editor.toolbar.focus-selected`	Frames the selection
Toggle Grid	`editor.toolbar.toggle-grid`	Shortcut G
Camera Path	`editor.toolbar.camera-path-edit|play|clear`	Author a fly-through for video export; unavailable while wall editing
Import ▸	`editor.toolbar.import-floor-plan`	Opens "Choose import flow"
Delete ▸	`editor.toolbar.remove-floor-plan`	Remove Floor Plan / Remove Background

Right of the toolbar sits a **Units** segmented control (Imperial | Metric) and a persistent keyboard legend.

10.2 Editor state — the actual Zustand store

Transcribed field-for-field from `editorStore` in the production bundle, defaults included.

```
{
  selectedObjectIds: [],
  isolateMode: false,  isolatedObjectIds: [],
  tool: "select",
  units: "ft",
  gridSize: 500,
  rotationSnap: 10,                 // degrees
  showGrid: true,  snapToGrid: false,  showGuides: false,
  shadowsEnabled: true,  glowEnabled: false,
  lightingPosition: { x: 5, y: 10, z: 5 },
  lightingPreset: "sunset",
  cameraMode: "perspective",
  linePlacement: { item: null, offsetMm: 0, angleDeg: 0, baseAxis: "auto" },
  shapePlacement: { kind: null }
}
```

Isolation is modelled carefully and worth copying: `toggleIsolation`, `extendIsolation` (add newly created objects to an active isolation), `pruneIsolation` (drop deleted ones, and exit automatically if the set empties) and `clearIsolation`. Angle helpers normalise to 0–360°, snap to the rotation increment, and format as `+90°`.

10.3 Camera and views

MODE	BEHAVIOUR
Top T	Orthographic. Correct for placement, spacing, selection box, floor-plan tracing. Some tools force a switch to it.
Perspective P	For height, realism, camera paths and client presentation.
Navigation	Orbit / pan / zoom, plus WASD fly movement.
View cube	Top-right of the canvas — a quick orientation control.
Axis gizmo	Bottom-right — RGB XYZ orientation indicator.
Camera Path	Ordered path points with per-path playback settings and animation speed; drives Export Video.

10.4 Selection and transforms

A floating bar sits above the canvas with **Move · Rotate · Scale · Delete**. On selection, a **radial context menu** appears at the object: cut, delete, pin/lock, duplicate, favourite, paste.

ACTION	SHORTCUT	NOTES
Move	G	Translate along gizmo axes
Rotate	R	Snapped to 10° by default
Scale	E	Supported object types only
Toggle snap	S	Grid snapping
Duplicate	`Shift+drag`	—
Delete	`Del` / `Backspace`	—
Deselect	`Esc`	—
Undo / Redo	`Ctrl/Cmd+Z` · `Ctrl+Shift+Z` / `Ctrl+Y`	—
Copy / Cut / Paste	standard	Via the radial menu too

Objects created by Table Designer carry a `group_id` (e.g. `group-table-group-7`) so a table, its linen, its chairs and every place setting move as one.

10.5 Scene lighting

- **Light Position** — a circular XY drag control with a numeric readout (e.g. `-6.8, 4.5, -1.8`)
- **Height (Z)** — slider
- **Shadows** and **Glow** checkboxes (Glow = `UnrealBloomPass`)
- **Lighting Preset** — twelve HDRI environments, shipped as 1k `.hdr` files:

UI LABEL	HDRI ASSET
Sunset (Venice)	`venice_sunset_1k.hdr` — the default
Dawn	`kiara_1_dawn_1k.hdr`
Night	`dikhololo_night_1k.hdr`
Forest	`forest_slope_1k.hdr`
Lebombo	`lebombo_1k.hdr`
Mountain	`immenstadter_horn_1k.hdr`
Bridge	`adams_place_bridge_1k.hdr`
Plaza	`potsdamer_platz_1k.hdr`
Outdoor (Park)	`rooitou_park_1k.hdr`
Warehouse	`empty_warehouse_01_1k.hdr`
Indoor	`st_fagans_interior_1k.hdr`
Studio	`studio_small_03_1k.hdr`

These are the standard Poly Haven HDRI names — all CC0 licensed, so Nozila can use the same set legitimately.

10.6 Save, autosave and history

- Explicit **Save** plus autosave after scene updates.
- Saved state preserves object transforms, textures, walls, floor plan and background references, lighting and camera.
- Leaving with unsaved work prompts: "We couldn't save your changes... leave the editor without saving your latest changes?" with a *Leave without saving* escape.
- Failure copy is reassuring and specific: "Your editor is still open and your current scene remains available. No navigation has occurred."
- Undo/redo is a bounded in-memory stack — it does not survive reload.

> **WORTH COPYING**
>
> The save-failure handling is unusually good. It never navigates away on failure, it tells the user exactly what is still safe, and it offers *Retry save*. For a tool where a lost scene is an hour of work, this is the difference between an annoyance and a churn event.

10.7 Admin-only performance overlay

With admin mode active, a small overlay appears at the top-left of the canvas showing **FPS**, **Scene tris** and **Selected tris**, labelled "Admin-only perf overlay". Cheap to build, and invaluable for diagnosing customer reports of slow scenes.

SECTION 11

Layout & geometry tools

Table Designer, Quick Layout, Line Placement, shapes, walls, rooms and openings.

11.1 Table Designer

The highest-value tool in the product. It generates an entire banquet arrangement in one pass, and keeps every generated clone in sync afterwards.

Configuration sequence

1. **Table Category** and **Table Shape** — round · rectangular · other
2. **Choose Table** — searchable picker; a segmented control switches between *Tables with Linens* and *Without Linens* catalogues
3. **Choose Linen** — separate picker; per-table linen variations, with "Use the table's default linen variation"
4. **Choose Chair** and **Number of Chairs** — "Seat count affects every generated table set."
5. **Place setting slots** — "Apply place settings per seat"
6. **Quick Layout** preset and spacing — "Enable the footer switch to configure layout presets and multi-table spacing."
7. **Number of Tables**, live **Total seats** readout
8. Live 3D preview, then **Insert**

Place setting slots

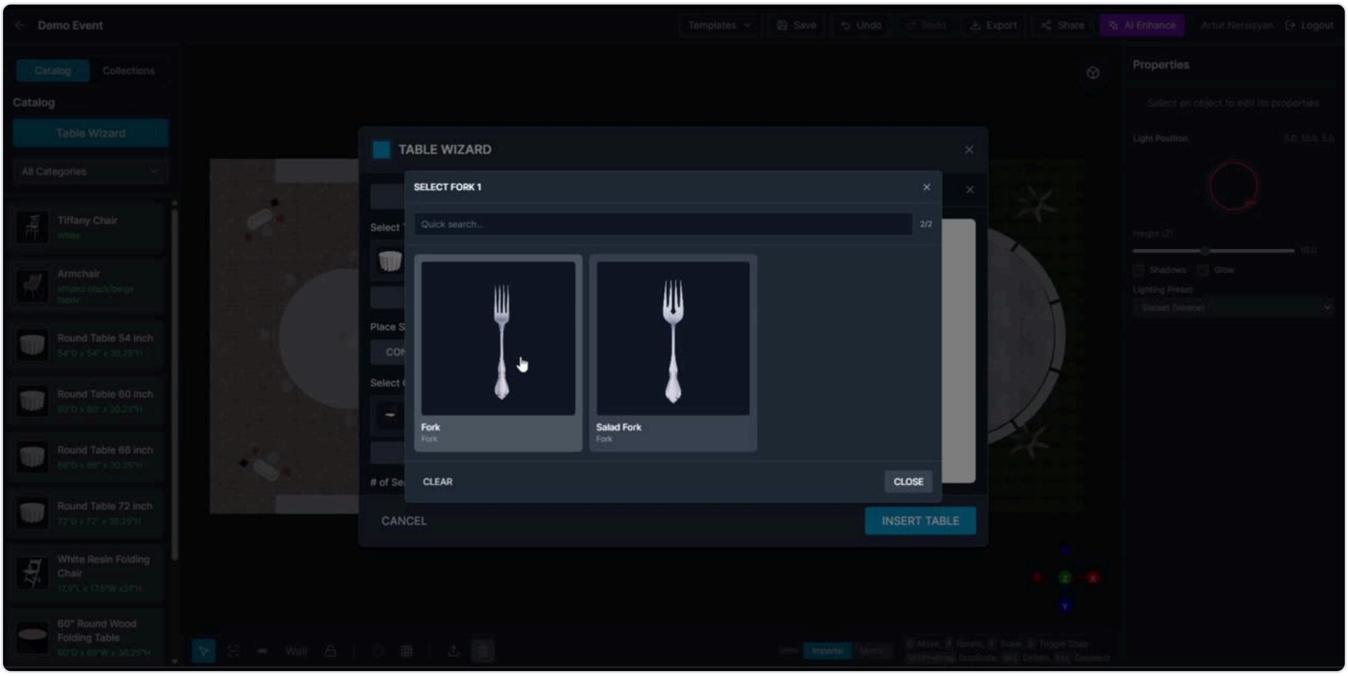
SLOT	HELPER TEXT OBSERVED IN THE PRODUCT
Plate / Platter	"Choose the primary plate used at every generated seat."
Beverage (Glass)	"Choose the beverage glassware placed with each seat."
Beverage, second cup	"Cup 2" / "Second glass"
Flatware — Fork 1	"Choose the utensils cloned onto each place setting."
Flatware — Fork 2	
Flatware — Knife	
Flatware — Spoon	
Centerpiece	"Choose the centerpiece placed at the middle of every generated table."

THE SYNCHRONISATION RULE — QUOTE IT INTO YOUR TICKET

"Every configured slot is cloned onto each generated seating position and **stays in sync with table, chair, and Quick Layout changes**."

That is the hard part. Changing the chair after generation must re-derive every seat position and move all eight place-setting items per seat. Implement place settings as a *derived* instanced group recomputed from the group's parameters — never as loose independent objects. The bundle confirms this with an `InstancedPlaceSettings` component.

Slot management affordances: *Clear Slot*, *Clear Section*, *Clear All*, and "Back to place setting slots". In applied mode, "Applied mode locks slot editing."



The nested picker pattern. A slot ("Select Fork 1") opens a sub-modal over the designer with quick search, a result count (`2/2`), a card grid, and Clear / Close. The parent modal stays mounted behind it. *(Earlier generation, named "Table Wizard".)*

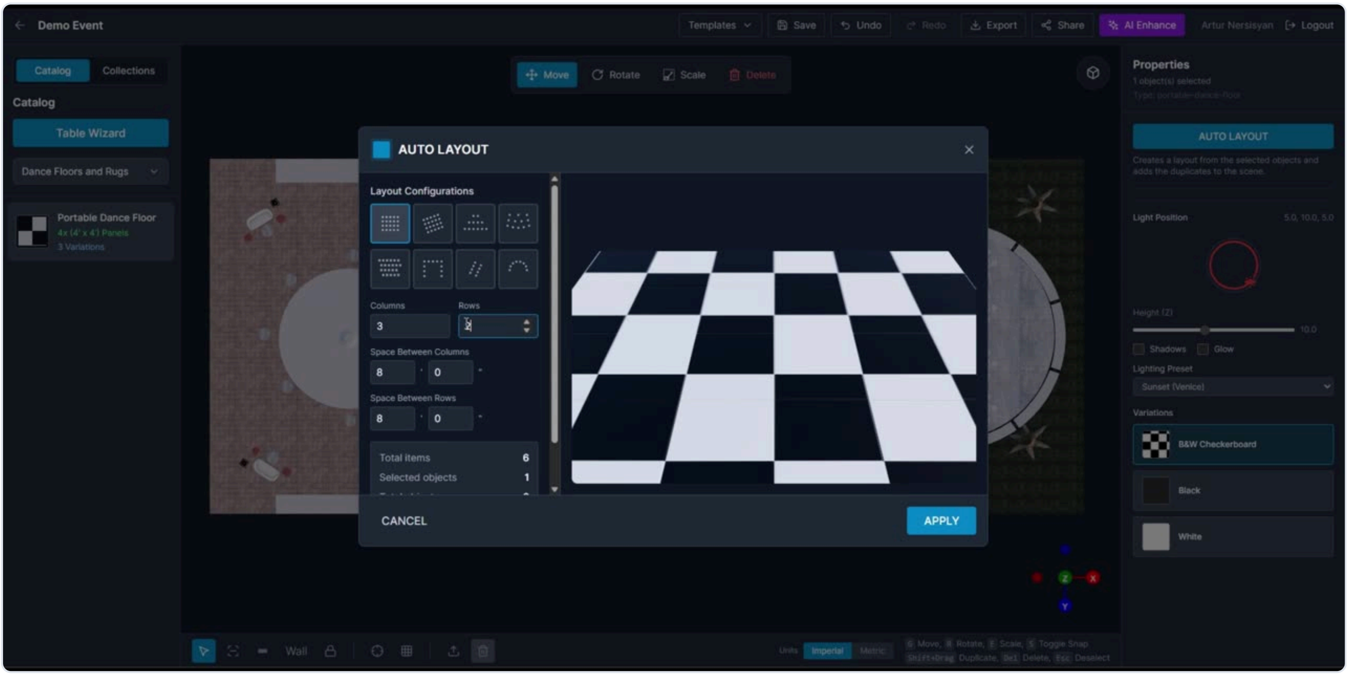
11.2 Quick Layout

Takes the current selection and generates a duplicated arrangement. Available both inside Table Designer and as a properties-panel action on any selection.

PRESET	USE
Grid	Rows × columns — the banquet default
Angled Grid	Rotated grid for diagonal rooms
Rows	Ceremony seating
Pyramid	Narrows toward one end
Diagonal	—
Curved	Arc facing a focal point
Circle / Closed Circle	Around a stage or centrepiece
U Shape	Conference and boardroom
Aisle	Splits with a centre aisle — "Items in First Row", "Items on Wings"

PARAMETERS

Number of Rows · Number of Columns · Items per Row · Space Between Rows · Space Between Columns (each as feet + inches in imperial) · Gap Between Items · Angle · Sweep (degrees) · Radius · Item increment. A summary shows **Total items**, **Selected objects** and total objects, with a live 3D preview.



Quick Layout. Eight preset thumbnails as a picker grid, numeric parameters below, a running totals block, and a live 3D preview filling the right half. Cancel / Apply. *(Earlier generation, named "Auto Layout".)*

11.3 Line Placement

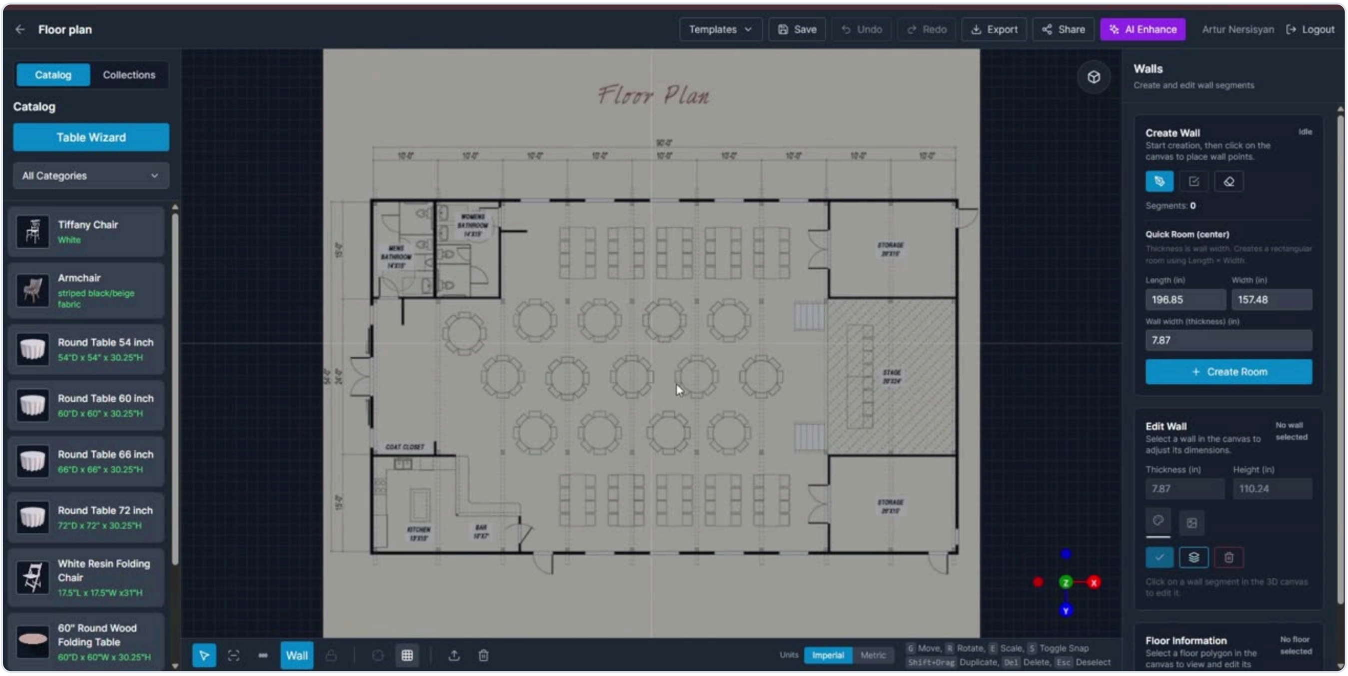
For chairs, stanchions, barricades and repeated decor. Configure item, spacing offset (`offsetMm`), placement angle (`angleDeg`) and base axis (`auto` | `X axis` | `Z axis`), then draw the run on the canvas — best in top view. Validation messages: "Enter a valid offset.", "Enter a valid angle."

11.4 Procedural shapes

Fast placeholder geometry for zones, platforms, scenic panels and measurement markers. Kinds: **Rectangle · Square · Circle · Triangle · Star · Box**.

Properties: Width · Depth · Height · Diameter · Radius · **Extrude Height** ("Default extrude is 0, which keeps the shape flat on the floor plane") · Fill Color · Fill Opacity · Border Color · Border Opacity · Shape color. Live readouts show Area, Center, Vertices and Segments.

11.5 Walls and rooms



Wall tracing over a calibrated plan. Right panel: *Create Wall* with an Idle/Drawing status chip and three icon actions (start / finish / clear), a live segment count, *Quick Room (center)* with Length, Width and Wall width inputs, then *Edit Wall* and *Floor Information*. The imported drawing sits dimmed, centred and locked on the floor plane.

Create Wall

- Click the pen icon (*Start creating*) → click points on the canvas → click the check (*Finish drawing*) or right-click to end a run.
- Status chip: *Idle* / *Drawing…*; live **Segments: n** counter.
- *Clear all walls* eraser action.
- Wall drawing is **3D only**. Shapes and camera path are disabled while wall editing is open.

Quick Room (center)

Enter **Length**, **Width** and **Wall width (thickness)** → *Create Room* generates a rectangular room at the origin. Helper: “Thickness is wall width. Creates a rectangular room using Length × Width.”

Edit Wall

Select a segment to edit **Thickness** and **Height**, apply a **Wall color** or **Wallpaper** texture, or delete the segment. Scope buttons make bulk edits fast: *Apply style to selected wall / Apply style to all walls*, and the same pair for floors.

Floor Information

Click a floor polygon to see Area, Center and Vertices, and apply a **Floor color** or **floor texture** with *Tile (m)* sizing and a tiling toggle.

Wall geometry persists as a `wall_blueprint` with spans (`span_id`, `parent_span_id`), segments (`segment_id`) and derived floor polygons.

11.6 Doors and windows as wall openings

Doors and windows are ordinary catalogue items that behave specially when dropped near a wall:

- The opening **snaps to the nearest eligible wall segment** and stores a `wall_attachment`.
- Editable properties: **width**, **height**, **bottom offset** (`bottom_mm`) and **side**.
- Top-view markers show opening locations.

- If an opening is near a wall end or taller than the wall, **the editor clamps it to valid limits** rather than rejecting it.
- Attachment data is saved with the blueprint so openings reload correctly.

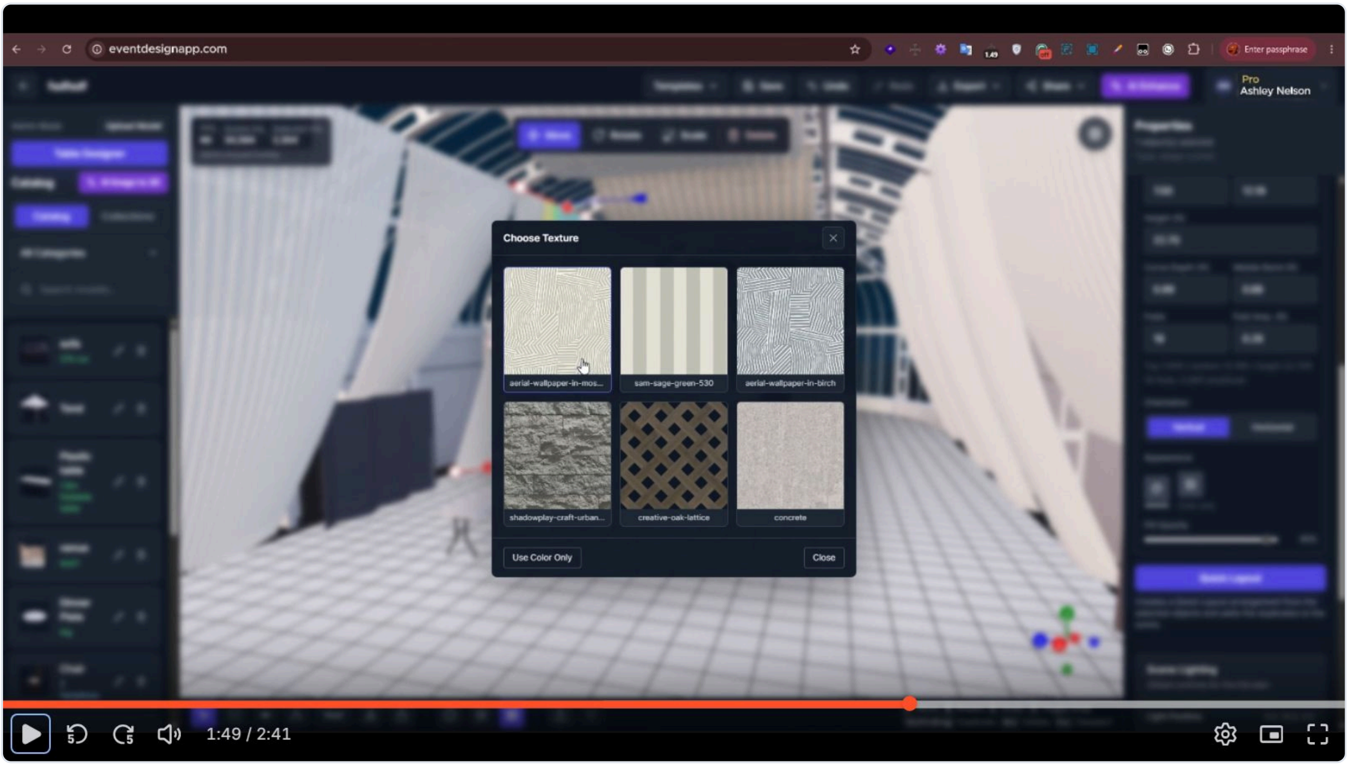
11.7 Materials, colours and textures

Per-material tinting

With exactly one catalogue model selected in 3D mode, a **Colors** control lists every material in the GLTF and allows a tint per material. "The tint is applied per configured material and preserves the original texture detail." Actions: *Apply*, *Reset all*, *Cancel*. Hidden for unsupported models, in 2D mode, and for multi-selection. Multi-colour is GLB/GLTF only. Stored as `material_color_config`.

Texture variations

Items may ship alternate textures selectable in the properties panel (e.g. a dance floor's *B&W Checkerboard / Black / White*). Uploading an item lets an admin attach variations, each requiring a name, a material id and a texture file.



Choose Texture. A simple swatch grid with names beneath, and a *Use Color Only* escape hatch that drops the texture in favour of a flat colour. The same picker serves walls, wallpaper, floors and linens, filtered by texture category.

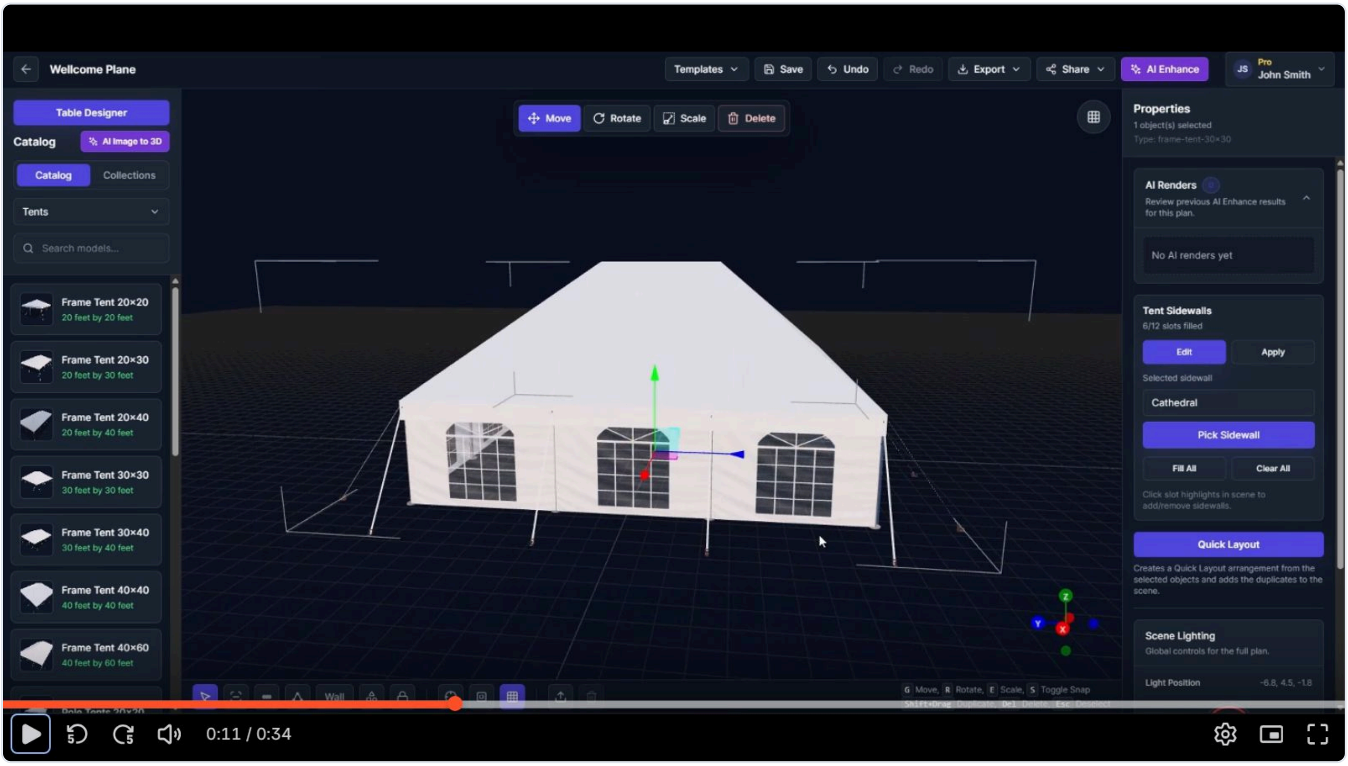
SECTION 12

The three builders

Tent, stage and drape — the domain-specific parametric objects that set the product apart.

These three are what separate the product from a generic furniture-placement tool. Each is a parametric object with a bespoke properties panel, and each encodes real trade knowledge. Budget accordingly — they are individually as much work as the whole catalogue system.

12.1 Tent builder



Tent selected, sidewall panel open. Right panel shows *Tent Sidewalls — 6/12 slots filled*, Edit/Apply, the selected sidewall type, a *Pick Sidewall* picker, and Fill All / Clear All. Left panel is filtered to the Tents category, each card giving true dimensions ("Frame Tent 20×30 — 20 feet by 30 feet").

Tent catalogue

Tents are catalogue items with a dedicated object type (`TentObject3D` , type slug `frame-tent-30x30`). Families and sizes observed:

FAMILY	SIZES SEEN IN THE CATALOGUE
Frame tents	20×20 · 20×30 · 20×40 · 30×30 · 30×40 · 40×40 · 40×60
Pole tents	20×20 and up
Sailcloth	Marketing-confirmed
High-peak marquee	Marketing-confirmed

The marketing site adds: "Need something not in the library? New tent types and custom dimensions can be created for your fleet." — i.e. bespoke tents are an onboarding service, not a self-serve feature.

Sidewall system

A tent exposes a fixed number of **sidewall slots** — one per bay per side — shown as a `filled/total` counter ("6/12 slots filled"). Slots are highlighted in the 3D scene and clicked to add or remove: "Click slot highlights in scene to add/remove sidewalls.”

- **Edit** enters slot-editing mode; **Apply** commits.
- **Pick Sidewall** opens a searchable picker of sidewall catalogue items.
- **Fill All** / **Clear All** for bulk work.
- Slots hold *different* wall types — the marketing framing is “open one face, glass on another, solid on the rest.”

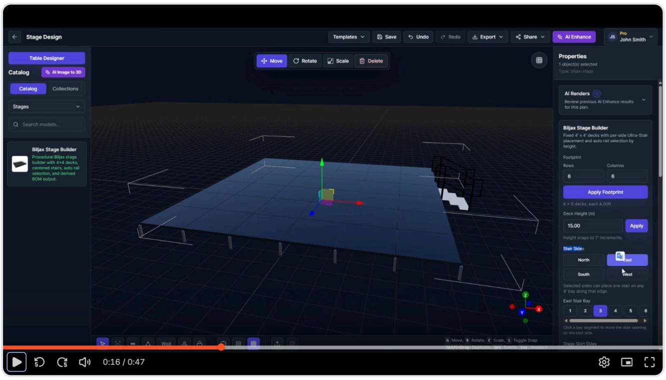
Sidewall types: Cathedral · Panoramic · Solid · Structured · Tinted windows · Arch windows · Curtains.

Data: `selected_sidewall`, `tent-sidewalls` slot array, `default-solid-panel` fallback. Empty state: “No tent sidewalls found in catalog.” Guard: “Select tent in scene to discover slots.”

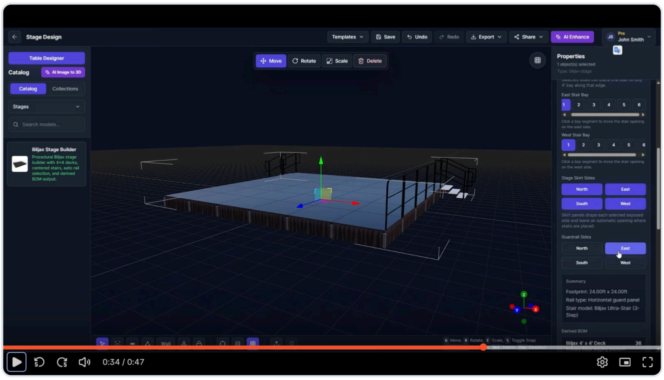
Interior design

The vinyl canopy can be **hidden** so the interior can be laid out and viewed. Everything else — tables, staging, draping, catalogue items — is placed inside at true scale as normal.

12.2 Biljax stage builder



Footprint, deck height, stair sides and per-side bay selection.



Skirt sides, guardrail sides, Summary block and the Derived BOM.

Presented as a catalogue item — “Biljax Stage Builder — Procedural Biljax stage builder with 4×4 decks, centered stairs, auto rail selection, and derived BOM output.” Object type `biljax-stage`.

Controls, in panel order

CONTROL	BEHAVIOUR AND EXACT HELPER TEXT
Footprint	*Rows* and *Columns* of 4′×4′ decks → *Apply Footprint*. Readout: “6 × 6 decks, each 4.00ft”.
Deck Height (in)	Numeric + *Apply*. “Height snaps to 1” increments.”
Stair Sides	North / East / South / West toggles. “Selected sides can place one stair on any 4′ bay along that edge.”
{Side} Stair Bay	A numbered segmented control (1…n) per selected side. “Click a bay segment to move the stair opening on the east side.”
Stage Skirt Sides	N/E/S/W. “Skirt panels drape each selected exposed side and leave an automatic opening where stairs are placed.”
Guardrail Sides	N/E/S/W. Rail *type* is chosen automatically by height.
Summary	“Footprint: 24.00ft × 24.00ft / Rail type: Horizontal guard panel / Stair model: Biljax Ultra-Stair (3-Step)”
Derived BOM	Live parts list — e.g. “Biljax 4′ × 4′ Deck … 36”

Parts catalogue

PART	DESCRIPTION AS SHIPPED
Biljax 4′ × 4′ Deck	“Primary Biljax staging platform deck.”

PART	DESCRIPTION AS SHIPPED
Biljax Stage Leg	—
Biljax Stair Support Leg	—
Biljax Snap Pin	“Snap pins for the primary stage legs.” / “…for stage legs and stair support legs.”
Biljax Rubber Base Pad	“Rubber base pads for the primary stage legs.”
Biljax Vertical Guardrail	Auto-selected above 30″
Biljax Horizontal Guard Panel	Auto-selected at or below 30″
Biljax Ultra-Stair	3-, 4-, 6-, 8- and 10-step variants
Stage Skirt 4′ Section	“Pleated black stage skirt section for a single exposed 4-foot bay.”

THE ENGINEERING RULES — VERBATIM FROM THE PRODUCT

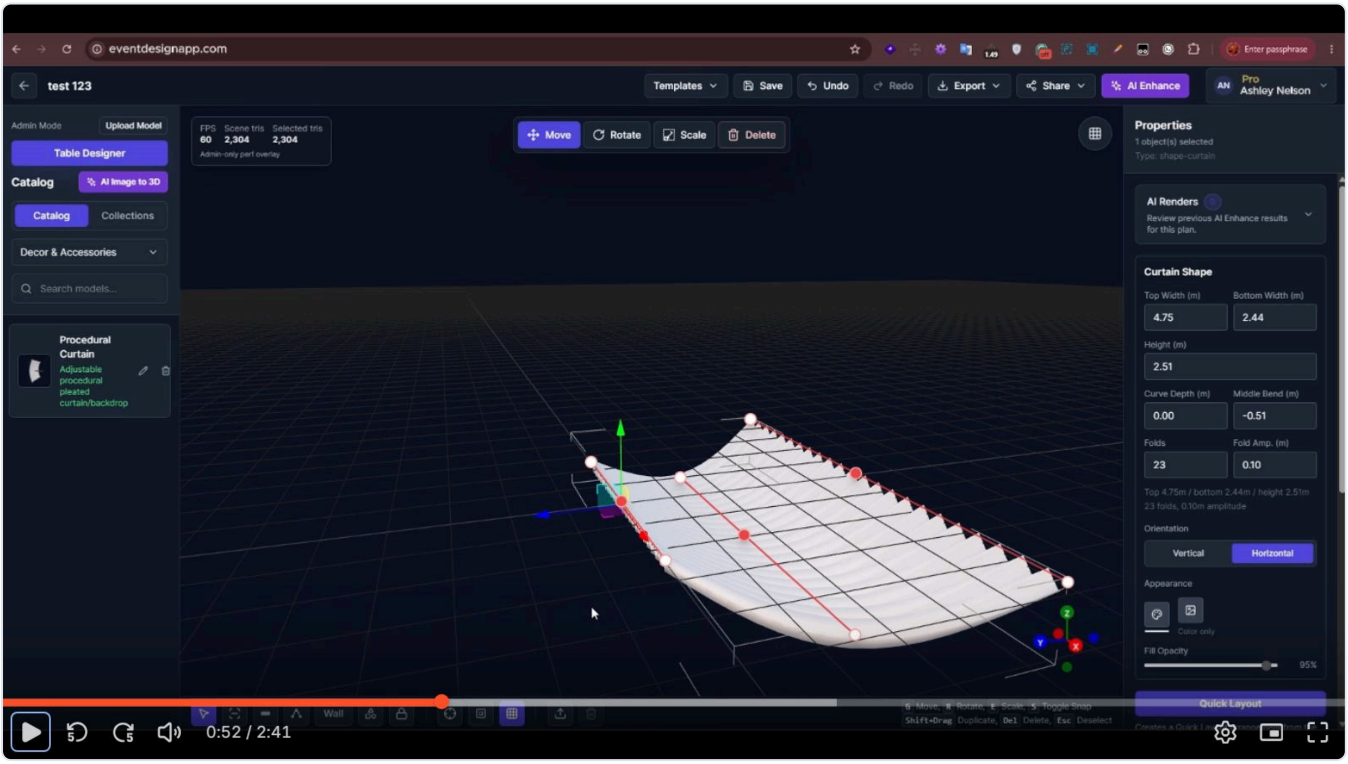
- “Auto-selected for stage heights above 30″.” / “…at or below 30″.” — guardrail type by deck height
- “Cross braces are required: brace each corner and every third bay with **2** braces per braced bay.”
- “Cross braces are required: brace each corner and every third bay with **3** braces per braced bay.” — a higher-height variant
- “Each stair requires **2 additional support legs and 2 additional snap pins** above 36″.”
- “Additional support legs required for stairs above 36″.”
- “Selected stair sides require a supported Biljax Ultra-Stair height between **12″ and 70″**.”
- “…on exposed edges. Add railing on: {sides}” / “Consider adding railing on: {sides}” — required vs advisory

COMMERCIAL AND SAFETY CAUTION FOR NOZILA

Biljax is a real manufacturer. Shipping their brand name, part names and load rules implies an endorsement the subject may or may not have. Two consequences for Nozila:

- Model the *behaviour* — modular decking, height-driven railing, bay-aligned stairs, derived BOM — under generic naming, or secure a manufacturer agreement before using their marks.
- Whatever the naming, put a visible disclaimer on BOM output. A parts list that a crew treats as engineering-approved is a liability surface, and the subject’s own terms already disclaim dimensional accuracy for exactly this reason.

12.3 Procedural curtain / draping



Procedural curtain. Nine white anchor handles on red guide lines let the fabric be shaped directly in the viewport, while the right panel exposes the same parameters numerically. Note the admin performance overlay top-left and the "Procedural Curtain — Adjustable procedural pleated curtain/backdrop" catalogue card.

Curtain Shape parameters

FIELD	EXAMPLE	NOTES
Top Width (m)	4.75	`top_width_mm`
Bottom Width (m)	2.44	`bottom_width_mm`
Middle Width (m)	—	`middle_width_mm`
Height (m)	2.51	—
Curve Depth (m)	0.00	`curve_depth_mm` — bow of the run in plan
Middle Bend (m)	-0.51	`middle_curve_mm` — signed centre bend
Folds	23	`fold_count` — pleat count
Fold Amp. (m)	0.10	`fold_amplitude_mm` — pleat depth
Orientation	Vertical | Horizontal	Segmented control
Appearance	texture | colour	Two icon options plus "Color only"
Fill Opacity	95%	Slider — enables sheer/translucent fabrics

A live summary restates the configuration: "Top 4.75m / bottom 2.44m / height 2.51m · 23 folds, 0.10m amplitude".

Direct manipulation — nine anchors

The curtain carries a 3×3 handle grid (`top/middle/bottom` × `left/centre/right`), stored as six extent fields plus three widths: `top_left_extent_mm` , `top_right_extent_mm` , `middle_left_extent_mm` , `middle_right_extent_mm` , `bottom_left_extent_mm` , `bottom_right_extent_mm` .

- Dragging a left or right handle changes that row’s width **symmetrically**.
- Middle-column handles adjust the top curve, bottom curve and centre bend; dragging the top row can also change height.
- **Double-click a non-middle handle** to enter single-point edit and move one side only. UI hint: “Single-point edit · Drag anchor · Esc to exit”.
- Minimum width and height limits prevent collapse to an unusable size.

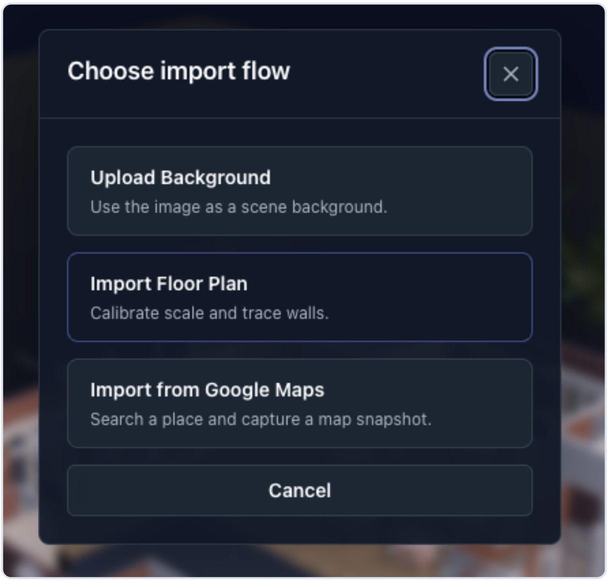
Use cases named by the product: pipe-and-drape, ceiling drapes, wall draping, ceremony backdrops, room dividers, fabric installations.

SECTION 13

Site context tools

Getting the real world into the scene: floor plans, backgrounds and aerial maps.

13.1 The import flow chooser



"Choose import flow" — three clearly differentiated options with one-line explanations. Reached from the upload icon in the bottom toolbar.

- **Upload Background** — "Use the image as a scene background."
- **Import Floor Plan** — "Calibrate scale and trace walls."
- **Import from Google Maps** — "Search a place and capture a map snapshot."

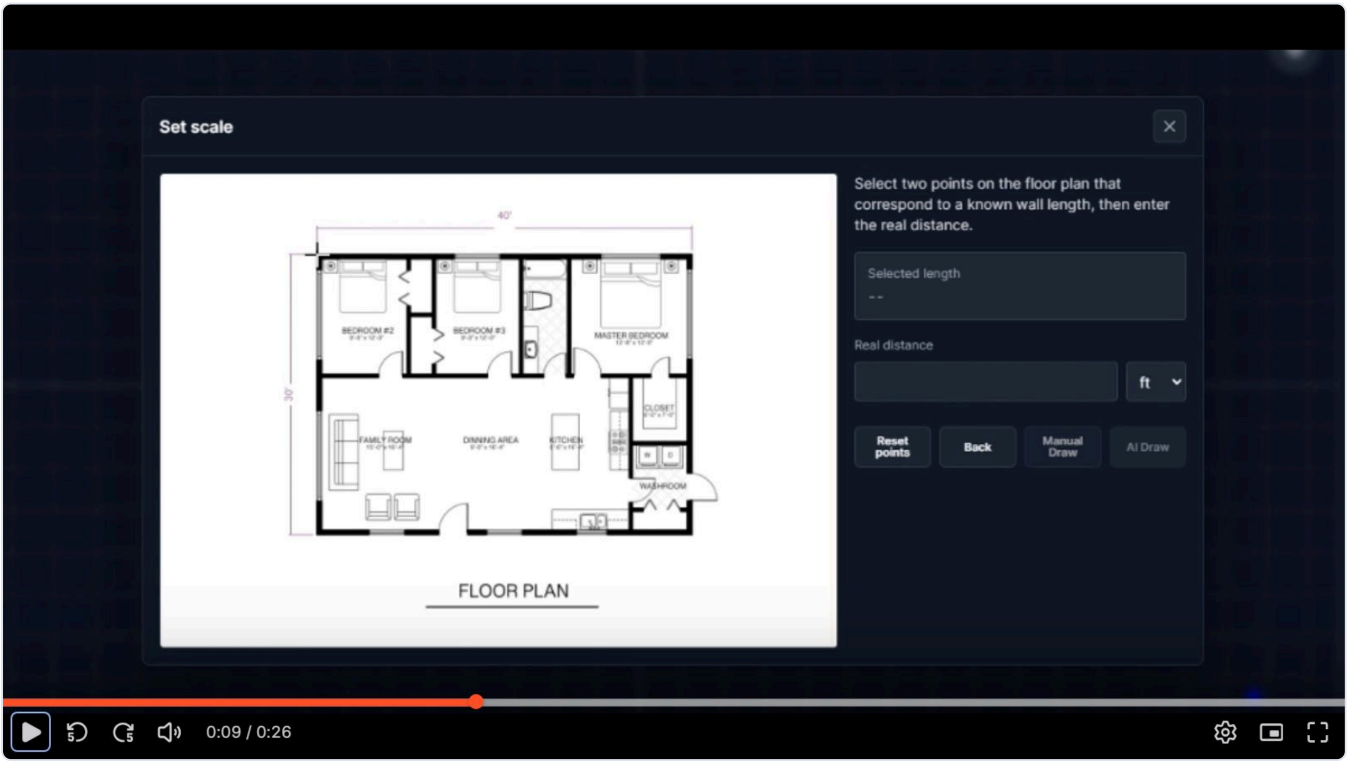
The import button is hidden entirely in view-only and shared guest links — an access decision, not just a disabled state.

13.2 Background image

Formats	PNG, JPG, WebP, GIF, SVG
Max size	10 MB
Placement	None — "the image is applied as soon as the upload finishes. There is no placement step."
Behaviour	Fills the canvas behind the scene in *both* 2D top view and 3D. Included in exports and renders. **Not** on the floor plane, and not moved, scaled, rotated or faded.
Replacement	Uploading a new background replaces the current one *and deletes the old file*.
Removal	Trash icon → Remove Background → confirm. Deletes the uploaded file too.
Gotcha	Isolate mode hides the background and the floor plan — a documented support issue.

13.3 Floor plan import and calibration

This is the feature that makes everything else trustworthy, and the help centre documents it in more detail than anything else in the product.



Two-point calibration. The drawing sits left; the right rail explains the task, shows the measured pixel distance ("Selected length"), takes the real distance with a unit select, and offers Reset points / Back / Manual Draw / AI Draw. Manual and AI Draw stay disabled until two points and a positive distance exist.

Step sequence

1. **Upload** — PNG, JPG, WebP, GIF or SVG; 10 MB max. Preview shown with a *Replace image* option.
2. **Set scale** — click one end of a known distance (a blue dot appears), then the other (a blue line joins them and the pixel distance is shown). Type the **Real distance** and pick a unit — m, cm, ft or in.
   - A third click restarts from a new first point; *Reset points* clears both.
   - Guidance worth reproducing: *use the longest measurement you are confident about, since calibrating off a short segment multiplies any click error across the whole plan.* A dimension line or printed scale bar is most reliable.
3. **Choose wall creation** — Manual Draw or AI Draw.
4. **Trace** — after Manual Draw the drawing is placed **dimmed, centred and locked** so it cannot be moved or selected by accident, and the view switches to Top.

Validation and error copy

- "Select two points to continue." and "Enter a valid distance." appear beneath the fields.
- "AI Draw supports PNG, JPG, WEBP, or GIF" — **SVG is rejected for AI Draw** (raster only); Manual Draw accepts it.
- "AI could not find usable wall segments." → the drawing is too faint, noisy or low-resolution.
- Scale is stored **per image**: a replacement drawing needs a fresh calibration.

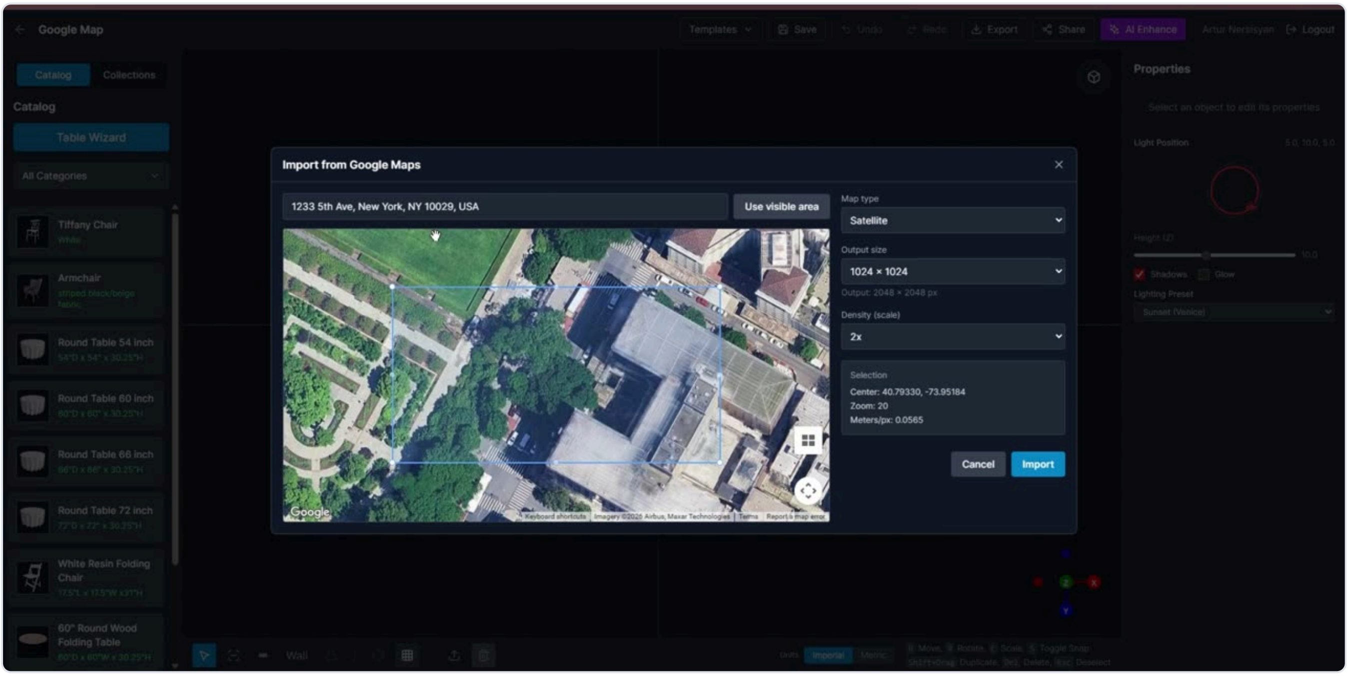
Manual Draw vs AI Draw

	MANUAL DRAW	AI DRAW
Availability	Free on every plan	Plus or Pro, and consumes credits (3)
Input	Any supported image incl. SVG	Raster only
Process	You trace with the wall tool	Async job with a status panel: "Starting AI Draw worker...", "Waiting for AI Draw worker...", "AI Draw running", "AI Draw complete"
Result	Exactly what you drew	"AI walls added. Review before saving."

	MANUAL DRAW	AI DRAW
Failure	—	Always falls back: “Continue with Manual Draw or try a clearer raster image.”

> **PATTERN TO COPY**
>
> Every AI feature here has a **free manual equivalent that the failure path routes to**. The AI is an accelerator, never a dependency. That single decision is why an AI provider outage degrades this product instead of breaking it.

13.4 Google Maps import



Aerial capture at true scale. Address search with autocomplete and a *Use visible area* shortcut, an interactive map with a selection rectangle, and a right rail for map type, output size, density and a live readout of centre coordinates, zoom and **metres per pixel** — the value that makes the capture measurable.

CONTROL	DETAIL
Place search	“Search a place or address” with Places autocomplete. Fallback: “Address suggestions are unavailable. Enter the address manually.” Guard: “Select a place that has location data.”
Use visible area	Captures the current viewport instead of a searched place
Map type	Satellite · Roadmap · Terrain (“Map”)
Output size	e.g. 1024 × 1024, with the effective pixel output shown (“Output: 2048 × 2048 px”)
Density (scale)	1× / 2× — the Static Maps scale multiplier
Selection readout	**Center** lat/lng · **Zoom** · **Meters/px**
Credits	1 credit, charged *only after a successful import*
Errors	“Failed to load Google Maps.” · “Invalid Google Maps response” · “Failed to import Google Maps image.” · “Map is not ready yet.”

The capture is performed server-side (`POST /uploads/google-maps-static`), which keeps the signing key private and lets the server compute and store `scale_world_per_px` so the image lands in the scene at true scale with no calibration step.

LICENSING NOTE

Google Maps Platform terms restrict how Static Maps imagery may be stored, modified and redistributed — and an event render sent to a client is redistribution. Check the current terms before shipping this, and budget for attribution being required on exports.

SECTION 14

The AI suite

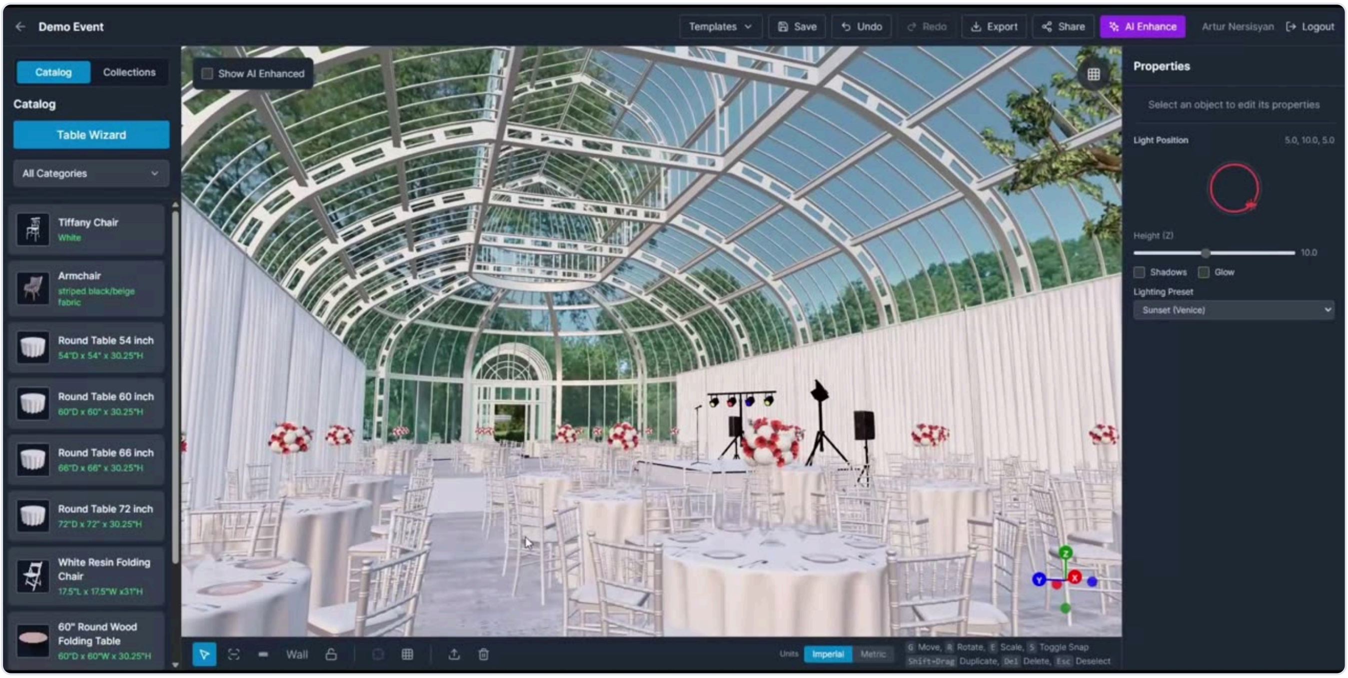
Four job types, one lifecycle. The commercial engine of the product.

14.1 The shared job lifecycle

```
┌─ preflight ─────────────────────────────────────────────────────────┐
│  plan tier?  company flag?  role?  credit balance?                   │
│  fail → PLAN_UPGRADE_REQUIRED | INSUFFICIENT_CREDITS | feature off   │
└───────────────────────┬──────────────────────────────────────────────┘
                        ▼
client generates  Idempotency-Key = "{action}-{uuid}"
                        ▼
POST → 202 { job_id | run_id, status: "queued" }          — credits debited
                        ▼
GET  status  ·  poll  ·  queued → in_progress(progress %) → completed | failed
                        ▼
┌ completed ──────────┐   ┌ failed / cancelled ───────────┐
│ artefact URL returned│   │ credits refunded · fall back to the │
│ appended to history  │   │ free manual path where one exists   │
└──────────────────────┘   └─────────────────────────────────────┘
```

All four report into one operational view (`/admin/reports/ai-process-runs`) with common columns: user, email, company, plan, provider, model, status, reference id, error, timestamps. Build the abstraction once.

14.2 AI Enhance — photoreal render from the current view



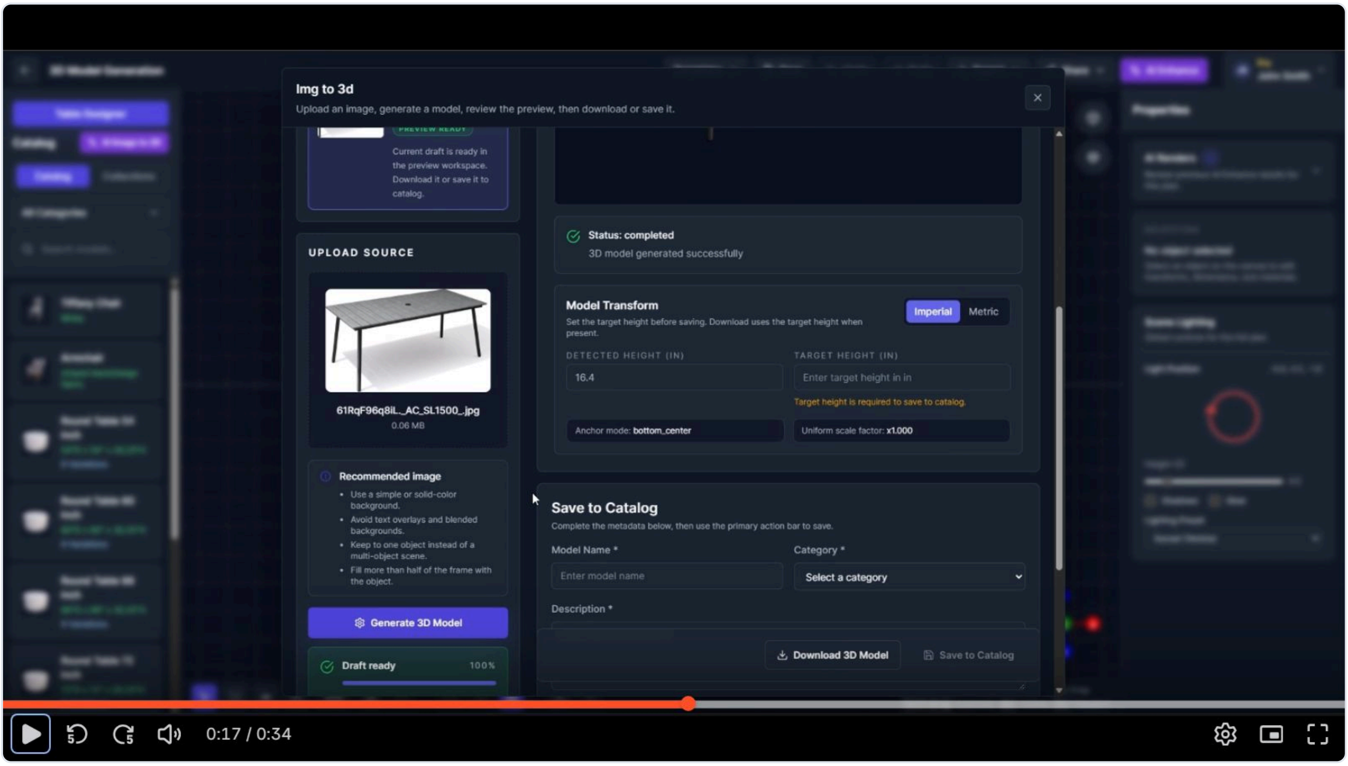
AI Enhance output. The result replaces the viewport image behind a *Show AI Enhanced* checkbox at the top-left of the canvas, so the user can toggle between the raw 3D scene and the render without leaving the editor.

What it does	Generates an improved image from the current plan view "while trying to preserve the existing structure, objects, camera angle, and layout."
Input	A canvas capture plus an editable **prompt** — "Describe how the image should be enhanced" / "Edit the enhancement prompt before generating the AI-enhanced image."
Models	A default model, a pro model (`nano_banana_2`) and a Qwen option — each with its own credit cost.
Cost	2 credits for the default; the pro and Qwen rates are admin-set.

Output	Toggled via *Show AI Enhanced*; appended to the plan's AI Renders history with a thumbnail.
History	"Review previous AI Enhance results for this plan." — collapsible panel, count badge, empty state "No AI renders yet".
Errors	"AI Enhancement failed", "Enhancement failed to start", "No canvas image available. Please ensure the plan is rendered."
Guest mode	Available to guests via a separate endpoint.

Marketing claims "<2 min render time". The states are staged carefully — *Enhance → Enhancing… → AI Enhancement complete* — with the header button itself carrying the progress.

14.3 AI Image to 3D — photo to catalogue asset



The Image-to-3D workspace. Three columns: source upload with guidance and a *Generate 3D Model* action; the live 3D preview with job status; and metadata for catalogue saving. The *Model Transform* block is the critical part — it converts an unscaled generated mesh into a correctly sized rental item.

Flow

1. **Upload** — JPG, PNG or WebP, 10 MB max.
2. **Configure** — model tier ("Pro 3.1", "Planner AI V1", "Planner AI V1 pro"), **PBR** toggle ("Use PBR mode for richer material output when needed"), optional Blender post-process.
3. **Generate** — 10 credits. Status: "Uploading image and creating worker job…" → "Starting worker…" → Queued → Working → *Status: completed — 3D model generated successfully*.
4. **Review** — orbit the preview; "Draft ready 100%".
5. **Set the height** — see below.
6. **Download GLB** or **Save to Catalog**.

Model Transform — the step that makes it usable

`source_height_cm`	"Detected Height" — measured from the generated mesh bounds
`target_height_cm`	Operator-entered real height. **Required to save to catalog**; must be > 0.

Uniform scale factor	Derived and displayed (`×1.000`)
`anchor_mode`	e.g. `bottom_center` — so the item sits on the floor rather than through it
Units	Imperial / Metric toggle within the modal
`transformed_glb`	The rescaled, re-anchored mesh that is actually stored

Helper text: "Set the target height before saving. Download uses the target height when present." · "Required for catalog save. Internally stored in cm."

Input guidance shown to the user

- Use a simple or solid-colour background.
- Avoid text overlays and blended backgrounds.
- Keep to one object instead of a multi-object scene.
- Fill more than half of the frame with the object.

Save to Catalog

Requires Model Name, Description and Category; **Scope** selects *My catalog*, *Company only* or *Global library* (the last two gated by role). A preview image is captured automatically from the 3D workspace. History is browsable and reopenable; already-saved runs are read-only.

Post-save advice from the help centre: check scale, texture quality and orientation before using generated models in production layouts.

The workspace also exists as a standalone route (`/dashboard/generate-model`), and can be disabled per account: "This workflow is currently turned off for this account. A super admin can re-enable it from the user settings in the Super Admin Console."

14.4 Floor Plan AI Draw

Covered end to end in §13.3. In job terms: input is the *calibrated* raster plan; output is a set of wall segments inserted into the scene for review; cost is 3 credits; failure always falls back to Manual Draw.

14.5 Venue Generator — photos to a navigable 3D environment

The most ambitious feature and the newest — labelled "Private, resumable generation · beta". It runs on its own route as a multi-step wizard rather than inside the editor.

Input modes

MODE	KEY	REQUIREMENTS
Single photo	`single`	"One clear photo of the venue." Sharp, well-lit, visible architecture.
Multi-view	`multi_view`	Front · 0° and Back · 180° **required**; Right · 90° and Left · 270° optional.
360° panorama	`panorama`	"One exact 2:1 equirectangular image." Projection, 2:1 dimensions and the left/right wrap seam are validated: "Width must equal exactly 2 × height after orientation normalization."

Upload limits: JPG, PNG or WebP · **10 MiB each, 40 MiB total** · all files uploaded together · **EXIF/GPS stripped**.

Wizard steps

1. **Prepare inputs** — choose the venue library or upload new photos. "Reuse an existing Venue from your library or upload a new venue photo."
2. **Analyzing inputs** — "Preparing and uploading the selected source images…"
3. **AI suggestions** — optional preset-driven cleanup proposals. Skippable: "Inputs uploaded. AI suggestions are unavailable, but you can continue with presets or the original set."

4. **Optional cleanup** — *one included request* per run. A single prompt ("Example: remove the bar stools and flower arrangements") is applied **atomically to every view**: "One prompt is applied atomically to every view; a partial result is never selected." Then choose *Use cleaned set* or *Use original set*.
5. **Generate Venue** — pick a model (`marble-1.0-draft`) and name it ("Example: Grand Ballroom"). Runs asynchronously: "World Labs is generating the Venue…"
6. **Venue ready** — SPZ Gaussian-splat preview; saved automatically to the reusable library "even if you close this window."

What the service guarantees

"The service preserves walls, doors, windows, columns, floor, ceiling and camera perspective." Cleanup is "one atomic, architecture-preserving edit across every input" that removes temporary items while preserving architecture.

Venue library and attachment

- Three scopes: **Public** (every signed-in planner), **Company**, **Personal**.
- Attach to a plan with *Use in this plan*. If one exists: "This plan already has a Venue. Replace only its environment? Furniture, groups and camera will remain unchanged."
- In the editor the venue can be locked/unlocked, hidden/shown, transformed, and detached. Detaching "removes the Venue environment from this plan… The reusable Venue stays in your Venue library."
- Quality modes: *adaptive*, *full* ("Load full-resolution Venue"), with graceful failure — "The Venue environment could not be loaded. Planner objects are still available."

Credits and failure handling

- "Every Venue model costs {n} credits"; charged when the run is **queued**.
- *Cancel generation · refund credits* is an explicit, labelled action.
- Provider billing failure is surfaced honestly: "World Labs could not start generation because its account needs active billing or available credits. Contact an administrator before retrying." (`provider_payment_required`)
- On failure: "Venue generation failed. Your selected inputs are still available." — inputs are never lost.

> **RECOMMENDATION FOR NOZILA**
> This feature carries the highest cost and the least proven value in the product: a 5 MB client renderer, an expensive third-party dependency, multi-minute latency, and a beta label. **Defer it.** A calibrated floor plan plus a photographic background delivers most of the client-facing benefit at a fraction of the cost. Revisit once the core product is earning.

SECTION 15

Output & distribution

Export, share, embed, templates, collections and the guest workflow.

15.1 Export

FORMAT	BEHAVIOUR
PDF	Printable plan review. **Requires the plan to be saved first** — "Please save the plan before exporting PDF". Company logo is used on PDF exports.
PNG	Quick image of the current view.
Video	Records a camera-path fly-through client-side via `MediaRecorder` as `video/webm;codecs=vp8`. Availability is browser-dependent: "Video export is not supported in this browser."

States are explicit throughout — *Exporting PDF... / PDF exported / PDF export failed*. The server returns a download URL; a missing one is reported as "Export response missing download URL." Emits an `export_completed` analytics event.

15.2 Share links

- *Share → Copy share link* mints a token and copies a read-only URL (`/share/:token`).
- Read-only plans cannot be shared: "Read-only plans cannot be shared."
- Viewer states: *Shared Plan Unavailable*, "This share link has expired.", "Share link not found.", "Missing share token".
- Shared views hide all editing affordances and show a "View only" badge.

15.3 Embed

Share → Embed iframe generates copy-paste iframe markup with configurable **Width** and **Height** (validated: "Use 100% or a pixel width.", "Use a pixel height."). Embedded views can remove the normal header entirely.

This is the mechanism behind the marketing claim "Let clients design their own events directly on your website" — a rental company embeds a plan or a guest session in its own site.

15.4 Templates

Purpose	Reuse a full plan setup — a room shell, a standard floor plan, a common seating arrangement.
Scopes	*Local* — "stay private to your account". *Global* — "available to every user."
Fields	Name (min 3 chars, unique — `TEMPLATE_TITLE_EXISTS`), description ("What this template is for"), visibility.
Loading	Destructive and warned: "Loading this template will remove current scene objects and replace them with template objects." Confirmation: *Replace Scene Objects?*
Overwrite	"Do you want to overwrite it with the current plan snapshot?" → *Overwrite Template?*
Constraint	"Please save the plan before creating a template."
Also used for	Onboarding starter workspaces — an admin maps each answer combination to a global template.

15.5 Collections

Reusable groups of selected objects — lounge sets, table groups, stage packages, decor clusters. Select objects → *Save as Collection* → name it. They appear under the Catalog panel's **Collections** tab and are placed like catalogue items.

- **Relative placement is preserved.**

- A preview image is captured from the 3D canvas at save time.
- Cards show name and object count ("Lounge — 5 objects", "Collection 1 — 1632 objects"), plus a sub-line listing contents ("Mic Stage Speaker and more").
- Guard: "Select one or more objects first." Empty state: "No collections yet. Save selected objects in the editor to create one."
- Drag-and-drop uses custom MIME types — `application/x-3d-catalog-item` and `application/x-3d-collection`.

15.6 Guest workflow

A no-account path intended for embedding on a rental company's own website.

1. **Guest Entry** — name, email, and a background photo (JPG/PNG, 10 MB max).
2. **Perspective extraction** — the app analyses the photo and derives camera parameters: **vanishing points**, **horizon line** (raw px and normalised), **FOV**, **tilt angle**, **yaw angle** and **camera position**. Progress: "Extracting Perspective..." → "✓ Perspective Extracted!"
3. **Guest Editor** — a reduced editor at `/guest-editor/:sessionId` with catalog and properties panels, AI Enhance and Export, but no save, share or import.

> **GENUINELY CLEVER, AND CHEAP TO BUILD**
>
> Perspective extraction means a guest's furniture sits in *their own photo* at a plausible angle instead of floating on a flat backdrop. It is classical computer vision — vanishing-point detection via a homography solve, which the bundle confirms ("Homography requires exactly 4 point correspondences") — not an AI call, so it costs nothing per use.
>
> Session state lives in browser storage, with an explicit quota-exceeded path ("Storage quota exceeded. Image will not be saved.") and periodic cleanup of stale sessions. Sessions expire: "The guest session could not be found or has expired."

SECTION 16

Plans, credits & billing

The commercial machinery — tiers, the credit economy, Stripe, and cancellation.

16.1 Onboarding and plan selection

New users land on a two-step onboarding before they ever reach the dashboard.

1. **"Tell Us About Yourself"**
   - *I am a…* — single select, required. "Choose one role to continue."
   - *I mainly design…* — multi select, at least one. "Select all that apply."

   Both option lists are admin-managed; the UI degrades honestly when empty ("No role options are available yet. You can still skip onboarding and start from the dashboard.").
2. **"Choose your plan"** — Free, Plus (badged *Most Popular*) or Pro, with live pricing and credit costs pulled from the API. Paid choices redirect to Stripe Checkout.
3. **Starter workspace** — "We will match a shared starter template from these onboarding answers and open a ready-to-edit *Wellcome plan* for you." (The misspelling is theirs, and appears in shipped plan titles.)

Draft answers are persisted under a local-storage key (`event-design:onboarding-checkout-draft`) so a Stripe round-trip does not lose them. Cancelled checkout returns with a clear message and the answers intact.

> SUPER ADMIN BYPASS, STATED PLAINLY IN THE UI
>
> "This account already has full access to all models and AI tools. No plan request, subscription approval, or credit gating applies to super admins."

16.2 The credit economy

Positioned to the user as: "Credits power the AI features. You only spend them when you run one — everything else is unlimited." This is a good frame and worth borrowing: it makes metering feel like fairness rather than restriction.

MECHANISM	BEHAVIOUR
Monthly grant	Plus 100 / Pro 200, granted at the start of each billing cycle. Admin-editable per tier; individual users can have a manual override ("Manual changes override the default plan quota and remaining credits for this user.").
Credit packs	Purchasable by Plus and Pro — 50, 100, 150. Each pack is a Stripe one-time Price. "Credit packs are available on Plus and Pro plans."
Expiry	Purchased credits **can expire** ("One month expiry"); the subscription modal shows when.
Consumption	Per-action, per the credit-ratio table (§3.4). Checked at run time, not at render time.
Refunds	On cancelled or failed jobs — explicitly labelled "Cancel generation · refund credits".
Insufficient	`INSUFFICIENT_CREDITS` → "Not enough credits remaining for this feature." / "This action needs {n}…" / "More credits required".
Company pool	Either per-user allocations or a single shared pool (§17).
Reporting	Usage by feature and by member; CSV export.

16.3 Stripe integration

OBJECT	USAGE
Recurring Price	One per tier for new checkouts. Editing prices in the admin console "creates new recurring Stripe Prices for new Plus and Pro checkout sessions" — existing subscribers keep their old price, and `historical_price_ids` is retained.
One-time Price	Per credit pack. "Credit amount and USD amount changes create a new one-time Stripe Price for future purchases."
Checkout Session	For plans, credit packs, seats and member credits — four distinct return paths.
Customer Portal	Separate user and company portals.
Seat subscription	Company seats live on one company subscription: "Seat changes are charged to the company card and kept on one company subscription."
Proration	Mid-cycle member upgrades show *Prorated amount*, *Due now*, *Final charge today*, and distinguish *Estimated locally* from *Stripe preview*.

Checkout return handling

On return from Stripe the app reads `session_id` and a status, then completes the session server-side and invalidates caches. Notice codes observed:

NOTICE	MEANING
`subscription_success` / `subscription_cancelled`	Plan checkout
`credit_success` / `credit_cancelled`	Credit pack purchase
`seat_success` / `seat_cancelled`	Company seat
`member_credit_success` / `member_credit_cancelled`	Credits bought for one member

Sync failures are handled without losing the payment: "Subscription checkout could not be synced. Refresh this page to retry." The literal placeholder `{CHECKOUT_SESSION_ID}` is guarded against — a nice detail that catches a common Stripe misconfiguration.

16.4 Manual upgrade requests

Alongside self-serve Stripe checkout there is an approval queue. A user can end up in `PLAN_REQUEST_PENDING` — "Your plan change request is pending approval." An admin approves or rejects from *Pending Plan Requests*. States: *Requested* · *Awaiting User* · *Approved* · *Rejected*. This supports invoiced enterprise deals and manual comping.

16.5 Cancellation

1. Open Subscription from the account menu.
2. Choose a **reason** from an admin-managed list — required. "Choose a reason before scheduling cancellation."
3. Some reasons force a note (`requires_note`): "Tell us what would have made the plan work better."
4. Cancellation is **scheduled at period end**, not immediate: "Your plan stays active until {date}." A *Cancel scheduled switch* / *Keep plan* action reverses it.
5. Feedback lands in the admin console under Cancellation Feedback: "Review who cancelled, their required reason, and any extra feedback."

WORTH COPYING WHOLESALE

A structured, admin-editable cancellation-reason taxonomy with conditional free-text, feeding a reviewable feedback queue, is a churn-analysis system most products never build. It costs very little and it is the only reliable way to learn why people leave.

16.6 Feature gating in practice

SITUATION	USER-FACING TREATMENT
Feature above tier	Visible but locked, badged “Available on Plus or Pro”; clicking opens upgrade options
Catalogue item above tier	Card renders in a locked style: “This catalog model is available on Plus and Pro plans.”
Free-plan items	Flagged `is_free_plan_available` — “Free users can place this item”
Insufficient credits	Blocked at run time with the exact requirement and a buy-credits path
Company disabled the feature	“This workflow is currently turned off for this account. A super admin can re-enable it…”
Free account, team action	“Free accounts can’t add team users. Upgrade to Plus or Pro to add users.”

SECTION 17

Company workspaces

Seats, roles, shared credit pools and the company admin surface.

17.1 Becoming a company

Either register with account type **Company** (which collects company name, address, country, postal code, phone and a main email), or convert later from the account menu — *Convert to Company*, "Create a company workspace and invite your team."

17.2 Company profile

Name (min 2 chars), address, city (min 2), country (2-letter code), ZIP/postal (2–20 chars, alphanumeric with spaces and hyphens), main email, phone (7–20 digits, with international dial-code selection), and a logo — PNG/JPG/WEBP, max 10 MB. The logo is "used on PDF exports and shared embeds. Images are resized to 512 px."

17.3 Team and seats

CONCEPT	DETAIL
Adding members	Two paths — **Direct add** ("The company card is charged before Member is created") or **invitation** ("The company card is charged before the invitation email is sent").
Seat plan	Each seat carries its own tier. "Paid company seat with standard monthly credits" (Plus) / "...with higher monthly credits" (Pro).
Roles	*Member* or *Admin*. "Team management is available to company Owners and Admins."
Visibility	*Own-only* or *Company-wide* — controls whether a member's projects are visible to the company.
Statuses	Invited · Active · Pending payment · Payment failed · Cancelled
Mid-cycle upgrade	"Move this member from Plus to Pro for the current billing cycle," with a proration preview.
Removal	"Company-owned projects created by this member transfer to the company owner." Seat renewal is cancelled at period end.
Deletion	Distinct and harsher: "This deletes the user's live account immediately."
Invitations	Expire; can be cancelled, already-accepted, or blocked by incomplete seat payment. Preview shows company, seat plan, role, visibility, invited email and expiry before acceptance.

17.4 Credit sharing — the two modes

Credits Per User

"Each team member receives their own credits." Credits can be bought for one member: "The selected pack is charged to the company card and added to this member account."

Shared Credit Pool

"All team members draw from the same pool." Members show "Draws from company pool". Packs go to the pool: "...added to the shared company pool."

Switching modes is deferred — *Set Credits Per User next cycle / Set Shared Credit Pool next cycle* — with one exception: "This will apply immediately when the first member is added." Ledger reasons `transfer_in` / `transfer_out` record movement between personal balances and the pool.

17.5 Company Dashboard

Two distinct surfaces exist for company managers:

Company Workspace (`/dashboard/company-admin`)

- **Users** — searchable list with status and role; select one to inspect
- **Projects** — that user's projects, openable read-only
- **Planners** — that user's plans, openable read-only
- **Projects Overview** — company-wide, with Created by / Last updated

"The dashboard only shows company data that your account is allowed to view."

Company settings sections

Profile · Team & Subscription · Credits · Billing — the last covering company expenses, usage by feature, usage by member, credit history with CSV export, and Stripe portal access. Some tenants are billed outside Stripe: "Billing for this account is managed by an administrator."

SECTION 18

Super Admin Console

Eleven surfaces at `/dashboard/adminconsole`. This is not optional infrastructure.

The console's own subtitle enumerates its scope: *"Companies, model and Venue reviews, textures, users, subscriptions, onboarding, reports, analytics, and theme."* Because pricing, credit costs, onboarding options and the palette are all server-driven, the application literally cannot be configured without it. Build it in phase one.

18.1 Section inventory

SECTION	CAPABILITIES
1 · Analytics	Dashboard of registered users, active users ("Active users on Plus or Pro plans"), plan split, and "% of active users". Drill into an individual user's event stream.
2 · Companies	Create, edit and delete tenants. Per-company toggles: *Global Models*, *Global Textures*, *company admin can manage assets*. "Manage tenant access, global catalog permissions, and company-level asset controls." View a company's users.
3 · Users	"Search, filter, impersonate, and manage platform users." Filter by company, role and status. Create users with a temporary password. Edit plan tier, monthly quota and current credits ("Quota and remaining credits must be non-negative whole numbers"). **Login as user** (impersonation), reset password, block/unblock, delete.
4 · Model Review	Approve or reject user-generated catalogue models before they become available.
5 · Venue Review	The same queue for AI-generated venues.
6 · Textures	"Upload, organize, and maintain global or company-scoped texture sets." Category management, PBR on/off, bulk upload with per-texture descriptions.
7 · Subscriptions	The largest section — six sub-tabs, detailed below.
8 · Onboarding	Manage Section 1 ("I am a") and Section 2 ("I mainly design") options with display order and active flags, then the **template assignment grid**: "Map every onboarding answer combination to a shared starter template." Guard: "Add at least one option to both onboarding sections to generate assignment rows."
9 · Reports	Five operational report types — model generations, venue generations, AI Enhance, floor-plan tracing and Maps imports. Filter by status, mode, company, provider and date range; search by user, email, company, project, venue, provider, model, run id, source image or job id. Shows Total Runs and per-run Process Detail.
10 · Theme	"Choose the global Qiso palette. Users still control Dark or Light mode in Account." Named presets, applied globally.
11 · Company Users	Cross-tenant drill-down into any user's projects and plans.

18.2 Subscriptions section, in detail

SUB-TAB	WHAT IT CONTROLS
Stripe Pricing	The recurring monthly price for new Plus and Pro checkouts. Warning shown: creating prices affects *new* sessions only.
Plan monthly credits	"Credits granted to Plus/Pro users at the start of a new billing cycle." "New defaults apply on the next billing cycle. Existing custom user quotas stay unchanged."
Credit packs	Editable pack slots — credit amount, USD amount, display order, active flag.
Credit ratios	"Edit the credit costs used when users run paid AI and import features." Note: "These values control credits deducted from user balances. Stripe billing prices remain separate."
Pending Plan Requests	"Approve or reject manual upgrades to the Plus and Pro plans."

SUB-TAB	WHAT IT CONTROLS
Paid Subscribers	"Paid subscriber health, cancellation setup, and feedback." Shows current plan, cycle end, credits used, scheduled cancellations.
Cancellation Reasons	"These are the required options users see before a Stripe cancellation is scheduled." Label, *Require note*, order, active.
Cancellation Feedback	Read-only review of who cancelled and why.

18.3 Console UI conventions

The console reuses a small, consistent component kit — worth mirroring, because it is what makes eleven sections buildable at reasonable cost:

- A collapsible left nav ("Collapse sidebar" / "Expand sidebar") with icon-only mode
- `AdminPanel` — section wrapper with title, description and a right-aligned action slot
- Standardised table container, head row and row classes
- Four notice tones — neutral, success, warning, error
- Icon action buttons in three tones — neutral, success, danger
- Simple *Previous / Page n / Next* pagination
- A single modal shell with a fixed `max-w-lg` and a footer button row
- Empty states as dashed-border cards, and a per-panel loading string for every fetch

SECURITY REQUIREMENTS FOR THE IMPERSONATION FEATURE

"Login as user" is genuinely useful for support and genuinely dangerous. If Nozila builds it:

- Write an immutable audit record — who impersonated whom, when, and for how long.
- Time-box the session and make it obvious in the UI that impersonation is active.
- Block impersonation of other admins.
- Consider requiring a stated reason, and notifying the impersonated user.

SECTION 19

Build plan

Phasing, sequencing risk, and what to cut.

19.1 Phases

#	PHASE	CONTENTS	SHAPE
0	**Foundations**	MySQL schema · auth (email + Google, verification, reset) · users, companies, roles · theme tokens and server-driven config · the admin console shell with Users, Companies and Theme · object storage · CI	Large
1	**Workspace**	Dashboard · projects · plans · plan CRUD · scene document read/write · save and autosave · account settings and unit preference	Medium
2	**Editor core**	r3f canvas · top/perspective cameras · grid and snapping · selection and box-select · transform gizmos · undo/redo · isolate, lock, focus · lighting with HDRI presets · properties panel framework	Very large
3	**Catalogue**	Categories and items · search and infinite scroll · GLTF loading and caching · drag-to-place · dimensions and scale correctness · texture variations · per-material tinting · admin upload	Large
4	**Layout tools**	Quick Layout presets · Line Placement · procedural shapes · **Table Designer with place settings**	Large
5	**Structure**	Wall drawing and blueprint · Quick Room · wall/floor styling · doors and windows as openings	Large
6	**Site context**	Background upload · floor-plan import with two-point calibration · Manual Draw · Google Maps capture	Medium
7	**Output**	PNG and PDF export · share tokens · iframe embed · templates · collections	Medium
8	**Commerce**	Tiers and gating · credit ledger · Stripe checkout and portal · credit packs · cancellation flow · admin pricing, quotas, ratios and packs	Large
9	**AI suite**	Job subsystem · AI Enhance · Image-to-3D with model transform · Floor Plan AI Draw · admin AI reports	Large
10	**Teams**	Company seats · invitations · roles and visibility · shared credit pool · company dashboard · credit history and CSV	Medium
11	**Domain builders**	Tent with sidewall slots · modular stage with derived BOM · procedural curtain with anchor handles	Large
12	**Differentiators**	**Quoting and inventory** · video export · guest mode with perspective extraction · onboarding survey and starter templates	Medium
—	**Deferred**	Venue Generator and Gaussian-splat rendering	—

19.2 Sequencing risks

RISK	MITIGATION
Units retrofit. Starting in floats or metres and moving to integer millimetres later touches every geometry path.	Fix the convention in phase 0. Write the conversion helpers before the first mesh.

RISK	MITIGATION
Scene schema churn. Saved plans become unreadable when the document shape changes.	`schema_version` from the first save, plus a migration runner. Non-negotiable.
Credits as a counter. Refunds, expiry and pool transfers arrive later and the counter cannot express them.	Ledger from the start, even while only one grant type exists.
Admin console deferred. Pricing and credit ratios end up hard-coded and every change becomes a deploy.	Ship Users, Companies, Theme and Credit Ratios in phase 0.
Table Designer underestimated. Treated as a modal, it is actually a live-synchronised generative system.	Model generated sets as parameterised groups from day one, never as loose objects.
Catalogue content. The tool is worthless without models, and the subject's library is its moat.	Budget for asset acquisition and Image-to-3D as a content-production tool, not just a customer feature.
AI provider volatility. Models are renamed and deprecated constantly.	Abstract behind a provider interface; keep the model id in `credit_ratio` so swaps are configuration.

19.3 What to confirm with an authenticated pass

Section 2 listed five gaps. Mapped to phases, they gate the following work — everything else can start now:

UNKNOWN	BLOCKS	CHEAPEST WAY TO CLOSE IT
Response body shapes	Phases 1–3	One authenticated session with the network tab open; 30 minutes
Dashboard / project page layout	Phase 1	Screenshots after login
Subscription modal composition	Phase 8	Screenshots on a Plus account
Real catalogue taxonomy and volume	Phase 3	Read the category dropdown and item counts
AI latency and failure rates	Phase 9	Run each of the four job types once and time them

OPEN OFFER

With account credentials — ideally a Pro account, and separately a company-admin and a super-admin account if they exist — a second pass would close all five gaps and let this document be extended with real response schemas, exact dashboard layouts and measured AI timings.

19.4 Recommended cuts for v1

- **Venue Generator** — highest cost, least proven, beta in the subject itself.
- **Video export** — browser-dependent, and PNG plus PDF cover the sales conversation.
- **Guest mode** — clever, but it serves a distribution strategy that only matters once customers exist.
- **Global template library and onboarding assignment grid** — ship a single starter template first.
- **Shared credit pool** — per-user credits are enough until teams are actually buying.

Conversely, do **not** cut: unit correctness, the credit ledger, save reliability, and the free manual fallback behind every AI feature. Those four are what make the product trustworthy.

APPENDIX A

Endpoint index

Alphabetical quick reference. Full detail in §9.

/admin
analytics/dashboard
analytics/registered-users
analytics/registered-users/{id}/events
catalog-reviews
catalog-reviews/{id}
catalog-reviews/{id}/approve
catalog-reviews/{id}/reject
companies
companies/{id}
companies/{id}/users
company/projects/{id}/plans
company/users
company/users/{id}/plans
company/users/{id}/projects
onboarding/section-1
onboarding/section-1/{id}
onboarding/section-2
onboarding/section-2/{id}
onboarding/template-assignments
onboarding/template-assignments/grid
onboarding/template-assignments/{id}
reports/ai-process-runs
reports/model-generations
reports/venue-generations
subscription-billing-prices
subscription-cancellation-reasons
subscription-cancellation-reasons/{id}
subscription-cancellations
subscription-credit-packs
subscription-credit-ratios
subscription-plan-credits
subscription-requests
subscription-requests/{id}/approve
subscription-requests/{id}/reject
subscription-users
theme
users
users/{id}
users/{id}/impersonate
users/{id}/reset-password

/analytics
events · identify · visitor

/articles
by-key/{key}

/auth
forgot · google · login · me
me/password · register · reset

/company
billing/customer-portal
billing/expenses
convert
credits/history
credits/history/export
invitations/accept
invitations/preview
logo
member-credit-checkout-sessions/{id}/complete
members
members/{id}
members/{id}/credits/checkout-session
members/{id}/upgrade
members/{id}/upgrade/preview
profile
seats
seats/preview
seats/checkout-sessions/{id}/complete
settings/credit-mode

/model-generation
history
history/{id}
save-to-catalog
status/{job_id}
upload-image-v3

/plans
/ · {id} · ?project_id=
share/{token}
{id}/ai-enhance
{id}/ai-enhance/{jobId}
{id}/ai-enhance/history
{id}/export
{id}/floor-plan/ai-draw
{id}/floor-plan/ai-draw/{jobId}
{id}/share
{id}/venue
{id}/venue/transform
guest/{sessionId}/ai-enhance

/privacy consent

/projects
/ · {id}
{id}/venue-generation-capabilities
{id}/venue-generation-runs
{id}/venues

/subscription

verify · verification/resend
onboarding/section-1
onboarding/section-2
onboarding/bootstrap-assigned-template

/catalog
categories
items
items/by-ids
items/{id}
items/{id}/model
items/{id}/texture-variations
items/upload-with-model
items/{id}/update-with-model

/collections
/ · {id}

cancel · cancellation-reasons
checkout-session
checkout-sessions/{id}/complete
checkout-sessions/{id}/complete-onboarding
credit-checkout-session
credit-checkout-sessions/{id}/complete
credit-ratios · customer-portal · pricing

/templates
/ · {id} · {id}/apply-to-plan

/textures
assets · assets/{id} · categories · sets

/theme config

/uploads
background · google-maps-static · {file_key}

/venue-generation-runs
{id} · {id}/cancel · {id}/finish
{id}/generate · {id}/image-revisions
{id}/suggestions · {id}/suggestions/skip

APPENDIX B

Interaction & analytics reference

B.1 Keyboard map

KEY	ACTION
G	Move (also Toggle Grid in the toolbar hint)
R	Rotate
E	Scale
S	Toggle snap
V	Select tool
T	Top view
P	Perspective view

KEY	ACTION
`Del` / `Backspace`	Delete selection
`Esc`	Deselect · exit single-point edit
`Shift+drag`	Duplicate
`Ctrl/Cmd+Z`	Undo
`Ctrl+Shift+Z` · `Ctrl+Y`	Redo
`WASD`	Fly navigation
Right-click	Finish a wall run

The editor renders this legend permanently at the bottom-right: "G Move, R Rotate, E Scale, S Toggle Snap / Shift+Drag Duplicate, Del Delete, Esc Deselect".

B.2 Analytics events observed

`page_view` · `registration_started` · `product_viewed` · `category_viewed` · `search_performed` · `export_completed` · `bounds_changed` · `credit_purchase` · `credit_success` · `credit_cancelled` · `credit_error` · `credit_mode_cycle_grant` · `credit_mode_transfer_in` · `credit_mode_transfer_out` · `subscription_success` · `subscription_cancelled` · `subscription_error` · `stripe_checkout` · `venue_generation` · `venue_dispatch_retrying` · `ai_enhance` · `ai_enhance_qwen` · `ai_enhance_nano_banana_2` · `ai_image_to_3d` · `floor_plan_ai_draw` · `google_maps`

`editor.toolbar.select` · `select-box` · `line-placement` · `shapes` · `wall-draw` · `lock-toggle` · `isolate` · `focus-selected` · `toggle-grid` · `camera-path-edit` · `camera-path-play` · `camera-path-clear` · `import-floor-plan` · `remove-floor-plan` | `editor.transform.move` · `rotate` · `scale` · `delete` | `editor.history.undo` · `redo` | `editor.view.toggle-top` | `editor.catalog.ai-image-to-3d`

APPENDIX C

Credit & feature-gate matrix

FEATURE	CREDITS	FREE	PLUS	PRO	ADDITIONAL GATES
Core editor, transforms, walls, shapes	`0`	YES	YES	YES	—
Starter model set	`0`	YES	YES	YES	Items flagged `is_free_plan_available`
Full catalogue library	`0`	—	YES	YES	Company `global_models_enabled`
Global textures	`0`	—	YES	YES	Company `global_textures_enabled`
Manual Draw (floor plan)	`0`	YES	YES	YES	Explicitly free on every plan
Background image upload	`0`	YES	YES	YES	Hidden in view-only and guest links
Export PDF / PNG	`0`	YES	YES	YES	Plan must be saved for PDF
Export Video	`0`	YES	YES	YES	Browser support; camera path required
Share link / embed	`0`	YES	YES	YES	Not on read-only plans
Templates & collections	`0`	YES	YES	YES	Global scope is admin-only
Google Maps import	`1`	—	YES	YES	Charged only on success; needs configured keys
AI Enhance (default)	`2`	—	YES	YES	Company enablement
AI Enhance (pro / Qwen)	`admin`	—	YES	YES	Model selection
Floor Plan AI Draw	`3`	—	YES	YES	Raster input only
AI Image to 3D	`10`	—	YES	YES	Per-account `ai_image_to_3d_enabled`
Venue Generation	`admin`	—	YES	YES	`venue_generation_access_override` ; provider billing
Save model to catalog	`0`	—	YES	YES	Company/global scope needs asset-management rights
Team seats	`$`	—	YES	YES	Company account; owner or admin role
Buy credit packs	`$`	—	YES	YES	Packs must be active
Admin console	`–`	SUPER ADMIN ONLY			Bypasses all plan and credit gating

APPENDIX D

Dependency register

Third-party services in the subject, with recommended positions for Nozila.

SERVICE	ROLE	POSITION	RATIONALE
Stripe	Billing	Adopt	Checkout, Portal, recurring and one-time Prices, seat subscriptions and proration are all exercised here. No credible alternative at this complexity.
Google OAuth	Sign-in	Adopt	Low cost, high conversion.
Google Maps	Aerial capture	Adopt with care	Keep the key server-side. Review Static Maps terms for storage and redistribution before shipping exports.
Sentry	Errors	Adopt	A 3D editor fails in ways logs alone will not explain.
Yandex Metrika	Analytics + replay	**Replace**	Session replay of a paid B2B tool is a privacy and procurement liability. PostHog gives the same insight with a defensible data story.
Gleap	Support, tours, KB	Adopt or substitute	The knowledge base is doing real work here — it is how users learn calibration. Whatever tool you pick, budget for writing ~50 articles.
World Labs	Venue generation	**Defer**	Beta, expensive, slow, heaviest client payload in the product.
Image-to-3D provider	Mesh generation	Adopt, abstracted	Fast-moving field. Keep model ids in configuration so providers can be swapped without a deploy.
Image-to-image model	AI Enhance	Adopt, abstracted	Same reasoning. Offering two tiers at different credit costs is a good pattern to copy.
Poly Haven HDRIs	Lighting	Adopt	CC0 — the same twelve environments can be used legitimately.
3D model library	Catalogue content	**Build / license**	The single largest non-engineering cost, and the subject's real moat. Do not treat it as an afterthought.

CLOSING NOTE

Everything in this document describes what exists today at eventdesignapp.com, evidenced from public assets and published documentation. It is a map of a competitor's decisions, not a licence to copy their assets, brand or code. The parts worth taking are the *decisions*: millimetre integers, a credit ledger, one job subsystem, a free manual path behind every AI feature, and an admin console that makes the product configurable without a deploy.

The part worth doing differently is the one they advertised and never built — turning a finished layout into a priced, signable quote.